

Two Routes to a Motor Prior: Guidance Versus a Frozen Body Model

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Abstract

Guidance and a body model are not interchangeable motor priors. We instantiate both under a single SAC backbone on MyoSuite musculoskeletal bodies and give each the adversarial control the other has. The first aid is a frozen body forward model, learned once from task-agnostic, reward-free motor babbling and used for short-horizon Dyna rollouts; those rollouts evaluate the environment’s analytic reward at the imagined states, so every reading of that arm is scoped to a forward model plus a known reward function. The second is a distilled coach, a frozen teacher policy acting through an auxiliary distillation loss. All main results use $n=12$ matched seeds and paired t 95% intervals. Guidance buys the early speed, by 50.6 mrad on myoElbow and 11.7 mm on myoFinger, and it is bought by what the teacher contains: a random-initialized teacher in the same loss does not reproduce that gain on myoFinger and is worse than no teacher at all on myoElbow. Cutting the coach is separately costly, at +16.0 mrad and +46.4 mm, though only the elbow cost reads as a loss of what the teacher knew. What the body model buys arrives later: the coach’s lead decays on both bodies and crosses zero inside the 12k budget on myoFinger but not on myoElbow. The babble-trained prior beats a random-initialized one on both bodies, on train targets. Both aids reduce myoElbow endpoint error while attached, by 20.7 and 31.6 mrad against model-free SAC; on myoFinger neither is shown to beat model-free RL at the endpoint. Four MyoSuite bodies were trained on and two carry the analysis. Under Holm–Bonferroni the myoFinger reversal survives four of the six family definitions we report and fails the unpruned literal set; seven cells survive every definition in which they appear, four of them adversarial controls.

1 Introduction

Humans and animals acquire motor skills in a handful of trials. Deep reinforcement learning on muscle-actuated bodies can require hundreds of millions of environment steps. Computational-sensorimotor theory offers two candidate explanations for the gap. First, the nervous system carries internal forward models of the body that predict the sensory consequences of motor commands (Wolpert et al., 1995; Kawato, 1999), and pairs those with inverse models indexed to objects and contexts (Wolpert & Kawato, 1998). Through experience the same system adapts an internal model of the dynamics it encounters, represented in the intrinsic coordinates of joints and muscles (Shadmehr & Mussa-Ivaldi, 1994). Second, the guidance hypothesis (Salmoni et al., 1984) holds that augmented feedback guides the learner toward the correct response while breeding a dependency on it, so performance measured *while* the aid is present can overstate what survives its removal. (§2 records how much of that hypothesis its own literature actually supports.)

We ask what each of these buys an RL agent under conditions where the two can be compared directly. We do not propose a new algorithm. We run a controlled study of two routes to a motor prior under one identical SAC backbone, on musculoskeletal bodies where the dominant lineage (MyoSuite baselines, SAR, DEP-RL, Lattice) is model-free with no learned-then-frozen world-model prior. The two routes are (1) a frozen body forward model, learned once from cheap, reward-free motor babbling and used for short-horizon Dyna rollouts (the *forward-model* route), and (2) a distilled coach, a frozen teacher policy that guides the student through a distillation auxiliary loss (the *guidance* route).

Crossing the two gives a 2×2 , $\{\text{no body prior, body prior}\} \times \{\text{no coach, coach}\}$, under one backbone, whose four constant-schedule endpoint cells are tabulated in §4.1, with two further arms that make the comparison interpretable: a random-initialized frozen model in the identical Dyna machinery, and a matched withdrawal in which each aid is cut at a common training step.

It is not obvious in advance that the guidance route should win the early race. The frozen body model supplies learned limb dynamics and short-horizon imagined rollouts, extra gradient signal and data augmentation early in training, exactly the regime where model-based methods classically deliver their largest sample-efficiency gains (Janner et al., 2019; Chua et al., 2018). A priori the prior had a credible mechanism to lead early. Yet we detect no early advantage for it under the primary estimator on either analysed body.

The contribution is a controlled dissection and does not propose a new method. Seven experimental controls make the two routes comparable: a random-init frozen model (§4.6) and its counterpart on the guidance route, a random-init frozen teacher in the identical distillation loss (§4.2), scored held-out where **randprior** is scored on train; clean-teacher retraining (§3.4); a non-annealed constant-weight coach (§3.2); a strict prior withdrawal with the imagined-transition buffer purged (§3.4); an abrupt-cut coach matched to that withdrawal in schedule shape (§3.2); and the combined arm run under the constant schedule (§4.7).

Three method commitments make the resulting contrasts interpretable. We use a direct paired within-axis contrast whenever two routes are compared, not an inference from two separate significance verdicts. We require a within-body, per-seed window $\times$ condition interaction test for any claim that an advantage grows or shrinks. And we hold the prior to a babbling anchor: its dynamics data are undirected, reward-free and task-agnostic.

The results support a single thesis: the two aids are not interchangeable. The coach is the faster of the two, but its lead shrinks while it is still attached, and removing it is separately costly. The prior’s advantage over the coach appears only late, and on myoFinger late enough to reverse the ordering by 12k, on a body where neither arm is shown to beat model-free RL. Neither aid survives being made uninformative: a random-initialized model loses to the babble-trained one on both bodies, on train targets, and a random-initialized teacher does not reproduce the coach’s early gain.

2 Related Work

Model-based RL and imagination. World Models (Ha & Schmidhuber, 2018) trains a controller inside a learned dream; the Dreamer series (Hafner et al., 2020; 2021; 2025) optimizes an actor–critic on imagined rollouts, updating that model online on data its own policy collects, which DreamerV3 states plainly (Hafner et al., 2025) and which holds even where model and policy gradients are kept separate within an update (Hafner et al., 2021). MBPO (Janner et al., 2019) co-learns model and policy and bounds compounding error with *short* rollouts branched from real data, the basis for our short-horizon Dyna. PILCO (Deisenroth & Rasmussen, 2011) established handful-of-trials model-based data-efficiency; PETS (Chua et al., 2018) refits between trials and plans by CEM with no learned policy; TD-MPC2 (Hansen et al., 2024) learns model, reward and value jointly. We group these on one axis only: all keep the dynamics model learning during task training rather than freezing a pretrained one. The sample-efficiency lineage (Janner et al., 2019; Chua et al., 2018) sets the expectation that model-based benefits are largest *early*, and on myoElbow we detect no early prior advantage (§4.2).

Self-modeling and babble-learned body models. The closest conceptual ancestor is robot self-modeling: Bongard et al. (Bongard et al., 2006) have a robot generate exploratory actions chosen to maximize disagreement among candidate models, infer a self-model of its own morphology, and synthesize behavior by re-planning rather than RL. Goal babbling (Rolf et al., 2010) learns an *inverse* model by *directed* exploration; we use *undirected* babbling to learn a *forward* model and keep it frozen. Plan2Explore (Sekar et al., 2020) learns a task-agnostic world model in a reward-free phase, and APV (Seo et al., 2022) pretrains action-free video representations before fine-tuning.

Offline model-based RL. MOREL (Kidambi et al., 2020), MOPO (Yu et al., 2020) and COMBO (Yu et al., 2021) freeze a pretrained model and optimize under it, as Plan2Explore’s zero-shot mode does (Sekar et al., 2020), but they train on logged interaction data and constrain the optimization by pessimism (Kidambi et al.,

2020; Yu et al., 2020) or conservative value regularization (Yu et al., 2021). We freeze a model learned from reward-free, task-agnostic babbling and use it optimistically. Our prior arm is also handed the environment’s analytic reward at imagined states (§3.5), task information that family learns from data, though MOREL likewise assumes a known reward function (Kidambi et al., 2020). Appendix M gives the finer distinctions.

Guidance, augmented feedback, and policy distillation. The guidance hypothesis (Salmoni et al., 1984) holds that augmented feedback guides the learner toward the correct response while breeding a dependency on it, so frequent feedback can flatter acquisition relative to retention. We state it that way rather than as the familiar "helps acquisition, hurts retention", because its best-known test supports only the retention half: Winstein & Schmidt (1990) report a reliable benefit of reduced feedback at *delayed* retention while acquisition was not reliably degraded, and the associated reversal effect is meta-analytically weak (Marschall et al., 2007). Appendix M gives both in detail. The hypothesis’s methodological core is the acquisition-versus-retention distinction: a measure taken *while* the aid is present measures acquisition, and only a test *after removal* measures learning. Our coach is a policy-distillation auxiliary, impersonal guidance and never social scaffolding, and we separate the two limbs by running it at constant weight (§3.2) *and* cutting both aids at a common step (§3.4, results in §4.4).

Muscle and musculoskeletal RL. Mainstream muscle RL is model-free: SAR (Berg et al., 2023) uses a muscle-synergy *action-space* prior, Lattice (Chiappa et al., 2023) perturbs the policy’s latent state to correlate exploration noise across time and actuators, and DEP-RL (Schumacher et al., 2023) injects a self-organizing exploration mechanism that periodically takes over control. Muscle actuation itself improves data-efficiency and robustness in stabilization tasks (Wochner et al., 2022). MyoSuite (Caggiano et al., 2022) provides the bodies; the published suite covers `myoElbow`, `myoFinger` and `myoHand`, and the `myoArm` model of Appendix F postdates it. The closest muscle work builds learned world models but co-learns them during skill training: MuscleVAE (Feng et al., 2023) interleaves world-model and controller training, and FreeMusco (Kim & Lee, 2025) calls the musculoskeletal model itself a "strong prior", meaning muscle physics and morphology, whereas ours is a task-agnostic forward model frozen before policy learning. MuscleVAE does freeze a world model within a later stage. The contrast is therefore with *when and on what* the model is learned; freezing as such is not the point at issue. We are not aware of prior work crossing a task-agnostic, reward-free, babble-learned frozen forward model with a distilled coach under one muscle-control backbone and then subjecting both to a matched presence/withdrawal dissection.

3 Method

3.1 Bodies, metrics, evaluation

Bodies. `myoElbowPose1D6M` (elbow; 1 DoF, 6 muscles) and `myoFingerReachRandom-v0` (index finger; 4 joints, 5 muscles, 3-D fingertip target at site `IFtip`) carry the analysis; a third body, `myoHand` (39 muscles), was also run in full, and a fourth, `myoArm` (`myoArmReachRandom-v0`, 34 muscles), was instantiated and trained on at a single seed. `myoHand` is reported in §4.10 and `myoArm` in Appendix F; neither carries a claim. Bodies are driven with raw MuJoCo on the validated MyoSuite models. The dynamics state is $\mathbf{s} = [\mathbf{q}, \dot{\mathbf{q}}, \text{muscle activation}]$.

Metrics and units. `myoElbow` error is a joint-angle error in milliradians (mrad); $1 \text{ mrad} \approx 0.0573^\circ$. `myoFinger` error is a 3-D Euclidean fingertip error in millimetres (mm). The two bodies are never compared in magnitude. Tables listing elbow and finger cells side by side are lists of separate per-body results.

Evaluation protocol. Actions are greedy, the start pose is fixed, each episode is 100 environment steps, and every condition is scored on the same 20 held-out targets unless a train-target block says otherwise. Appendix I gives the protocol and the SAC hyperparameters in full.

Backbone. One SAC backbone, identical in every condition (Appendix I).

Real data per update. The number of gradient steps per environment step is identical in every condition, but the content of those steps is not. Each update in `prior`, `randprior` and `priorcoach` samples 64 real + 64 imagined transitions, while each update in `blank` and `coach` samples 128 real, at the same update-to-data ratio: the model-based arms receive half the real-data gradient signal per update. This disclosure has one

schedule	implementation	weight at 0 / 2k / 4k / 6k / 8k- 12k	what its endpoint measures
Tapered ("annealed")	<code>coach_coef = coach0-max(0, 1 - step/8000)</code>	1.00 / 0.75 / 0.50 / 0.25 / 0	a <i>fade</i> , scored 4k steps after the fade completes — not a matched withdrawal
Constant	<code>coach_anneal = 1e8</code>	1 — step/ 10^8 (1.00000 → 0.99988 over the budget)	performance under an aid that is never removed
Abrupt cut	weight held at <code>coach0</code> until 8k, then 0	1.00 / 1.00 / 1.00 / 1.00 / 0	retention , matched to the prior’s step-function removal

Table 1: The three coach schedules: the implementation of each, the distillation weight it carries at env-steps 0, 2k, 4k, 6k and 8k, and what its endpoint therefore measures.

exception, and §3.4 gives it: after a strict prior withdrawal the arm reverts to 128 real per update, so from 8k it is not receiving half but all of it. The matched control that would separate this from the model’s contribution, replacing the imagined half of each batch with replayed real transitions, was not run.

3.2 Conditions and coach schedules

blank is model-free SAC. **prior** adds the frozen babble model in a short-horizon Dyna loop. **coach** adds the distillation auxiliary loss with no body model, a frozen teacher policy distilled into the student through a loss pulling the student’s actions toward the teacher’s on visited states. **priorcoach** is both. **randprior** replaces the babble-learned model with a random-initialized frozen one in identical machinery, and **randcoach** is its mirror on the other route: a random-initialized frozen *teacher* of identical architecture, distilled through the same auxiliary loss at the same constant weight, evaluated held-out on the same twelve seeds as `coach(const)`. Withdrawal conditions cut one aid at env-step 8000 (§3.4). The coach encodes *what to do*. The body prior encodes *how the body moves*.

Only the abrupt-cut coach is a matched withdrawal: the constant and tapered arms already differ before the nominal removal point on elbow (`coach(const) - coach(anneal)`, EARLY, -10.76 $[-15.09, -6.42]$), whereas the prior arm’s withdrawn-minus-kept difference is exactly 0.000 at 2k, 4k and 6k. The constant arm is implemented at a weight just under 1.0 (0.99998 at 2k, 0.99988 at 12k). Measured against the true step-function abrupt arm over the same 12 seeds, the pre-cut difference (**abrupt** — **constant**, in the §3.6 direction) is $+1.65$ mm $[-5.35, +8.65]$ at 2k, $+1.67$ at 4k and $+1.27$ at 6k on finger, and ≤ 0.02 mrad at every pre-cut checkpoint on elbow, negligible beside the $+46.37$ mm withdrawal cost measured on the same pair.

3.3 The babble prior

The prior is an MLP $(\mathbf{s}, \mathbf{a}) \rightarrow \Delta \mathbf{s}$ trained on task-agnostic random (Ornstein–Uhlenbeck) muscle activations with no task and no reward, then reused frozen. Five-step open-loop accuracy is 19.3 mrad on myoElbow and 14.3 mm mean / 9.8 mm median on myoFinger. The archived build transcripts give 2×10^5 transitions for myoFinger and 3×10^5 for myoHand. On myoElbow no transcript survives, but the artefact does. `prior_my0.pt` is on disk, is loaded by every elbow **prior**, **randprior** and **priorcoach** run, and is bit-identical to the model the committed budget sweep produces at 10^5 transitions, so the elbow budget is 10^5 and the three bodies run 10^5 , 2×10^5 and 3×10^5 . A reproduction is not a log, and we report it as the weaker thing it is, what a committed script at its own defaults yields today rather than a record of what was run. The checkpoint’s accuracy is recoverable the same way. Re-measured under the protocol Appendix I states, it gives 19.3 mrad, reproducing the figure this paper has carried, and the same protocol on the on-policy state distribution, where the model is actually queried and where no fidelity figure in this paper had previously been measured, gives 24.9 mrad at five steps and 10.4 mrad at the three-step horizon the Dyna loop uses, against 10.9 mrad on babble at that horizon. The myoFinger and myoHand figures are still babble-distribution measurements and remain an optimistic bound. Appendix I gives the build details.

3.4 Teachers and withdrawal

Teachers are trained from **blank** for 30,000 rewarded, task-specific environment steps that no other arm receives, one teacher per body at a single seed, saved as a terminal checkpoint off a non-monotone curve with no model selection. Every teacher-dependent claim is therefore scoped to the teacher as captured. Held-out teacher scores are 27.2 mrad on elbow, far better than any student; on finger 169.1 mm, worse than every student condition mean bar one, the withdrawn finger coach (169.14) being numerically identical to it and just under the no-op referent of 172.3 mm. Appendix I gives the training details.

Each aid is cut at env-step 8000 and every condition is scored at 12000. The abrupt-cut coach and the two prior withdrawals are cut at a common step. The two prior withdrawals are bit-identical to their kept control before it; the abrupt coach differs from the constant coach by ≤ 0.02 mrad on elbow and up to 1.67 mm on finger pre-cut (§3.2). None of the three is a clean single-factor manipulation, and the strict prior withdrawal is not matched to the coach’s in what it leaves the agent training on (Appendix I.1): the strict arm also doubles its real-data gradient signal at the cut, and the soft arm keeps the mix on a stale buffer (§6). Appendix I gives the mechanics.

3.5 Budgets and the information each route receives

Neither intervention arm is matched to **blank**, and the two routes are not matched to each other. The prior arm consumes task-agnostic, reward-free babble transitions of the least task-relevant kind before its 12k-step run, $6.7\times$ the teacher’s 30,000 steps on myoFinger and $3.3\times$ on myoElbow, while the coach arm inherits 30,000 rewarded on-task steps. Appendix I gives the budgets.

There is a second asymmetry, in the opposite direction. `dyna_rollout` computes `r = task_reward(s2, target, ...)` on model-predicted states: the reward is a deterministic analytic function of state, evaluated at states the agent has never visited. The prior arm (and `randprior`, and both `priorcoach` arms) therefore receives a reward oracle evaluable at imagined states, task information no model-free arm has. Every biological or theoretical reading of the forward-model arm is accordingly scoped to "a forward model plus a known analytic reward function", never to a forward model alone.

3.6 Windows, contrasts and statistics

Six evaluation checkpoints at env-steps 2k, 4k, 6k, 8k, 10k, 12k. **EARLY** = mean over {2k,4k,6k}; **MID** = mean over {6k,8k,10k}; **LATE** = mean over {10k,12k}; **ENDPOINT** = the value at 12k, which is an endpoint at a fixed budget rather than a verified asymptote. **EARLY** and **LATE** are disjoint, which is what makes the window $\times$ condition interaction test well-posed. **MID** overlaps both, and **ENDPOINT** $\subset$ **LATE**, so a late-window result and an endpoint result count once between them, not twice. One training seed is the unit of analysis. Conditions share matched seed ids (seed0...seed11), every contrast is a paired difference over matched seeds with $n=12$ ($n=6$ on myoHand), and difference-of-difference contrasts are formed within seed before any interval is computed. Bodies are never pooled inside a contrast. Every contrast is written $A - B$ on reach error, so a negative value means A has lower error.

The primary statistic is the paired t 95% CI, $\text{mean}(d) \pm t_{0.975, n-1} \cdot \text{sd}(d)/\sqrt{n}$, reported as an effect size with its interval; a contrast is significant only when that interval excludes zero. Two robustness checks accompany it, a BCa bootstrap and an exact Wilcoxon, and neither licenses a claim the primary estimator does not: where they disagree we report the disagreement and do not choose among them. Contrasts are paired within seed, bodies are never pooled inside a contrast, and any claim that an advantage grows or shrinks requires a within-body per-seed window $\times$ condition interaction test. Multiplicity is handled by Holm–Bonferroni over six declared family definitions (§4.11). Appendix I states the rule in full, including the exclusion criterion that generates the window families and the ~ 0.1 -unit rounding allowance.

4 Results

All conditions are $n=12$ seeds, paired, held-out targets unless stated. Elbow numbers are mrad, finger numbers are mm. Negative = the first-named condition has lower error. Intervals are paired t 95% CIs. The

condition	elbow (mrad)	SD	finger (mm)	SD
blank (model-free SAC)	70.73	22.78	117.60	43.03
prior kept	50.02	8.18	90.50	25.57
prior soft-withdrawn	52.11	7.34	88.12	22.14
prior strict-withdrawn	57.19	26.05	104.69	31.98
coach constant	39.13	7.77	122.77	12.18
coach annealed (tapered)	61.07	30.66	135.15	46.10
coach abrupt-withdrawn	55.13	15.06	169.14	34.52
prior + constant coach	38.49	4.77	116.13	9.19
prior + annealed coach	57.72	13.98	114.49	17.84
randcoach (random teacher, constant)	93.26	21.63	142.02	17.97

Table 2: Held-out endpoint error at 12k for every condition on both bodies, with the across-seed SD. Elbow is mrad, finger is mm; n=12 matched seeds.

held-out endpoint (12k)	no coach	coach (constant weight)
no body prior	blank 70.73 / 117.60	coach 39.13 / 122.77
body prior	prior 50.02 / 90.50	priorcoach 38.49 / 116.13

Table 3: The 2×2 read off Table 2: held-out endpoint at 12k under {no body prior, body prior} × {no coach, constant-weight coach}, elbow mrad before finger mm.

subsections are ordered by the argument instead of by experiment number: §4.2 tests the thesis’s first part (guidance buys early speed) and §4.3 the endpoint comparison on which the route-versus-route claim rests, §4.4 its second (that cutting the coach is separately costly on both bodies, significantly so on each, and on myoElbow that cost significantly erodes the coach’s own endpoint advantage, and that we cannot detect a difference between the two routes’ removal costs), §4.5 and §4.6 its third (what the model buys arrives late, and only an informative model buys it). §4.7–4.10 are secondary observations and scope, and §4.11 is a robustness check on multiplicity.

4.1 Endpoint error by condition on both bodies

Table 2 gives the held-out endpoint means at 12k with their across-seed SD.

The four constant-schedule cells of §1’s 2×2, {no body prior, body prior} × {no coach, coach}, read off Table 2 and given in Table 3, elbow mrad first and finger mm second.

The two bodies are separate results in separate units and the cell entries are never compared across the slash. The same four cells under the annealed schedule, on both target sets, are plotted in Appendix E ([fig1_2x2_endpoint.png](#)).

Train-target endpoints, used by §4.6: blank 65.6 / 123.3; prior 40.1 / 89.0; randprior 89.4 / 129.8. The constant coach (39.13) and prior + constant coach (38.49) are the study’s lowest elbow held-out errors, both below the prior (50.02) and far below model-free SAC (70.73). On finger the two prior arms are the numerically lowest conditions, soft-withdrawn 88.12 and kept 90.50, but neither is distinguishable from model-free **blank** at 117.60 (soft-withdrawn – **blank** –29.49 [–67.41, +8.43]; kept – **blank** –27.10), so no finger condition is shown to beat model-free RL at the endpoint, the constant coach included. Figure 1 plots the held-out learning curves the table summarises.

4.2 The early window

`coach(const)` – `prior` in the EARLY window {2,4,6k}, held-out, is –50.56 on elbow and –11.72 on finger: the constant-weight coach is ahead of the frozen prior early on both analysed bodies. That lead is bought by what the teacher contains. The adversarial control for this route, **randcoach**, a random-initialized teacher of identical architecture in the same loss at the same weight, does not reproduce that

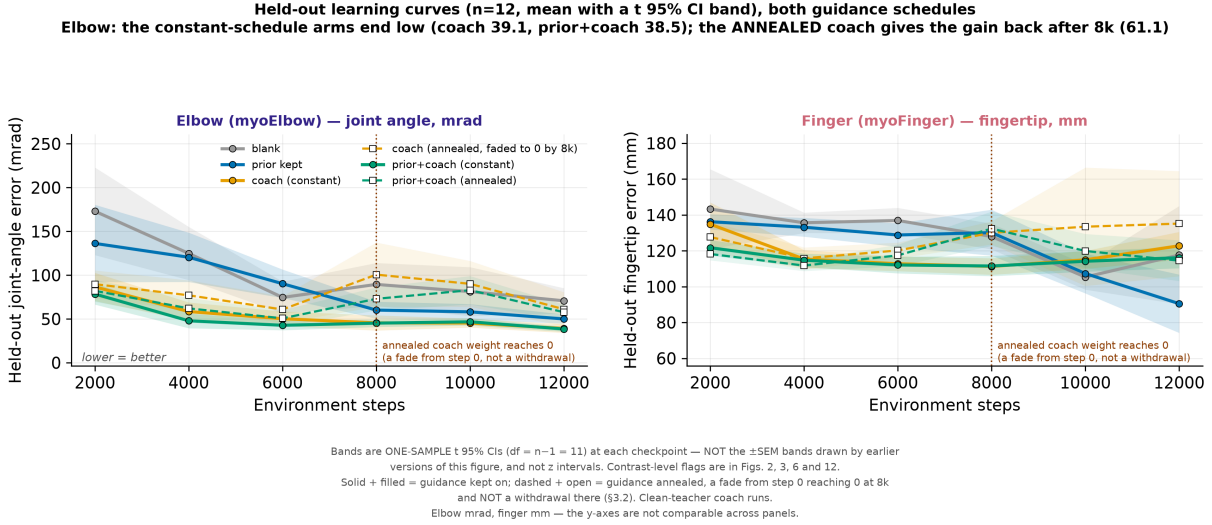


Figure 1: Held-out reach error against environment steps for each condition and body, with one-sample t 95% CI bands over 12 seeds. Elbow panels are mrad, finger panels are mm. The two share no scale.

contrast	body	mean	paired t 95% CI
prior — blank (held-out)	elbow	-20.72	[-35.85, -5.59]
prior — blank (train)	elbow	-25.50	[-39.03, -11.97]
prior — blank (held-out)	finger	-27.10	[-66.32, +12.12]
coach(const) — blank	elbow	-31.61	[-48.58, -14.63]
coach(const) — blank	finger	+5.17	[-26.49, +36.82]
coach(const) — prior	elbow	-10.89	[-19.26, -2.52]
coach(anneal) — blank (train)	elbow	-10.26	[-36.42, +15.90]
coach(anneal) — prior (train)	elbow	+15.24	[-8.32, +38.80]

Table 4: Held-out endpoint contrasts at 12k. Negative means the first-named condition has lower error.

gain on myoFinger and is significantly worse than no teacher on myoElbow. The direct within-axis contrast `coach(const) — randcoach` is -19.03 [-24.40, -13.67] on finger and -108.60 [-142.01, -75.18] on elbow; against model-free RL the same arm is +1.43 [-6.03, +8.89] on finger EARLY and +24.41 [-3.16, +51.99] at the endpoint, both failures to detect; the direct within-axis contrast at the endpoint is `coach(const) — randcoach` = -19.25 [-32.66, -5.84]. No `randcoach` contrast appears in RECOMPUTED.md; they are computed by `analyze_randcoach.py` under the primary estimator of §3.6 (Appendix I). §5 records why this control is the one the guidance claim needed. Against model-free RL in the same window the coach runs -59.03 on elbow and -17.60 on finger, and the tapered coach reproduces it (-48.27, -17.39). The prior’s own EARLY cell is not significant under the primary estimator on either body (elbow -8.47; finger -5.88, BCa-only, Appendix A), and it decomposes the elbow route contrast exactly: -59.03 + 8.47 = -50.56, so on elbow `coach — prior` is `coach — blank` plus a quantity we cannot distinguish from zero. This was not the a-priori favourite: the frozen model supplies exactly what classically pays off earliest (Janner et al., 2019; Chua et al., 2018), and the two EARLY nulls are where that expectation fails (§1). Appendix K carries the intervals.

4.3 The endpoint, with both aids attached

Table 4 gives the held-out endpoint contrasts at 12k.

The prior’s elbow effect is $70.7 \rightarrow 50.0$ mrad held-out, -20.7 mrad $\approx -1.19^\circ$, a ~29% reduction in reach error, significant on held-out and on train targets. The coach’s endpoint advantage has no train-target twin,

contrast	body	mean	paired t 95% CI
coach abrupt-withdrawn – coach const	elbow	+16.00	[+6.49, +25.52]
coach abrupt-withdrawn – coach const	finger	+46.37	[+22.88, +69.87]
prior strict-withdrawn – prior kept	elbow	+7.17	[−10.95, +25.29]
prior strict-withdrawn – prior kept	finger	+14.19	[−0.95, +29.33]
coach cost – prior cost	elbow	+8.83	[−13.33, +31.00]
coach cost – prior cost	finger	+32.18	[−3.02, +67.39]
coach abrupt-withdrawn – blank	finger	+51.54	[+12.63, +90.45]
coach abrupt-withdrawn – blank	elbow	−15.60	[−32.75, +1.54]
prior strict-withdrawn – blank	elbow	−13.54	[−35.98, +8.89]
prior strict-withdrawn – blank	finger	−12.91	[−52.54, +26.72]
coach-withdrawn – prior-withdrawn	finger	+64.45	[+31.76, +97.14]
coach-withdrawn – prior-withdrawn	elbow	−2.06	[−23.19, +19.07]
prior soft-withdrawn – blank	elbow	−18.62	[−31.25, −5.99]

Table 5: Withdrawal at 8k, scored at 12k. Contrasts are withdrawn minus kept, so positive means withdrawal made the condition worse.

because the constant schedule was never run there; the two schedules that were run do not beat the prior on train targets and are post-withdrawal numbers, not a still-attached twin. We report that asymmetry rather than leaving the prior’s corroboration row to stand alone. The constant-weight coach improves elbow endpoint accuracy by more (−31.6 against model-free SAC), and the direct route-versus-route contrast, the quantity a claim about which aid is stronger actually requires, is −10.89 in the coach’s favour: while present, the coach is the stronger aid at the myoElbow held-out endpoint. Nothing is claimed for the prior on finger, where the prior-versus-blank cells are significant under BCa but not under the primary estimator (Appendix A) and the MID and LATE cells under neither (Appendix D). Two scope conditions travel with these cells (Appendix K).

4.4 Withdrawal

The natural story after §4.3 would be that the coach is a crutch that works while you lean on it, whereas the body prior teaches the agent something it keeps, a story this subsection tests and does not support. We ran the control that story requires, with each aid cut at 8k and every condition scored at 12k. Costs are **withdrawn – kept**, so positive means withdrawal made the condition worse.

Cutting the coach is costly on both bodies. On myoElbow, where the teacher as captured is competent, that cost is a within-route acquisition-versus-retention dissociation and is the cleanest result in the withdrawal experiment; §5 reads the finger cost differently, since the finger teacher sits essentially at the no-op referent. On finger, removing the coach leaves the agent worse than never having had one (+51.54 against **blank**) and worse than the withdrawn prior (+64.45), both produced by the withdrawn finger coach landing essentially at the no-op referent rather than by the withdrawn prior being retentive. Cutting the prior is not shown to be costly on either body and no route-level durability asymmetry is claimed: the prior’s own withdrawal cost is non-significant on both bodies (elbow +7.17; finger +14.19), and so is the direct route comparison (elbow +8.83; finger +32.18). Neither null is evidence of retention, and the first has a named candidate generator that biases this contrast toward zero (§3.4); the finger interval misses zero by 0.95 mm. The cost does not identify two things: cutting the coach at 8k removes the teacher’s information *and* deletes a weight- ≈ 1.0 auxiliary loss term mid-training, and Appendix O reads the finger cost as dominated by that optimization shock. Neither prior withdrawal is single-factor either (§6). Appendix K carries the bias analysis and the soft-withdrawal cell, and Figure 2 plots the per-route withdrawal costs.

4.5 The time course of the coach’s lead

Table 6 gives `coach(const) – prior` held-out, by window.

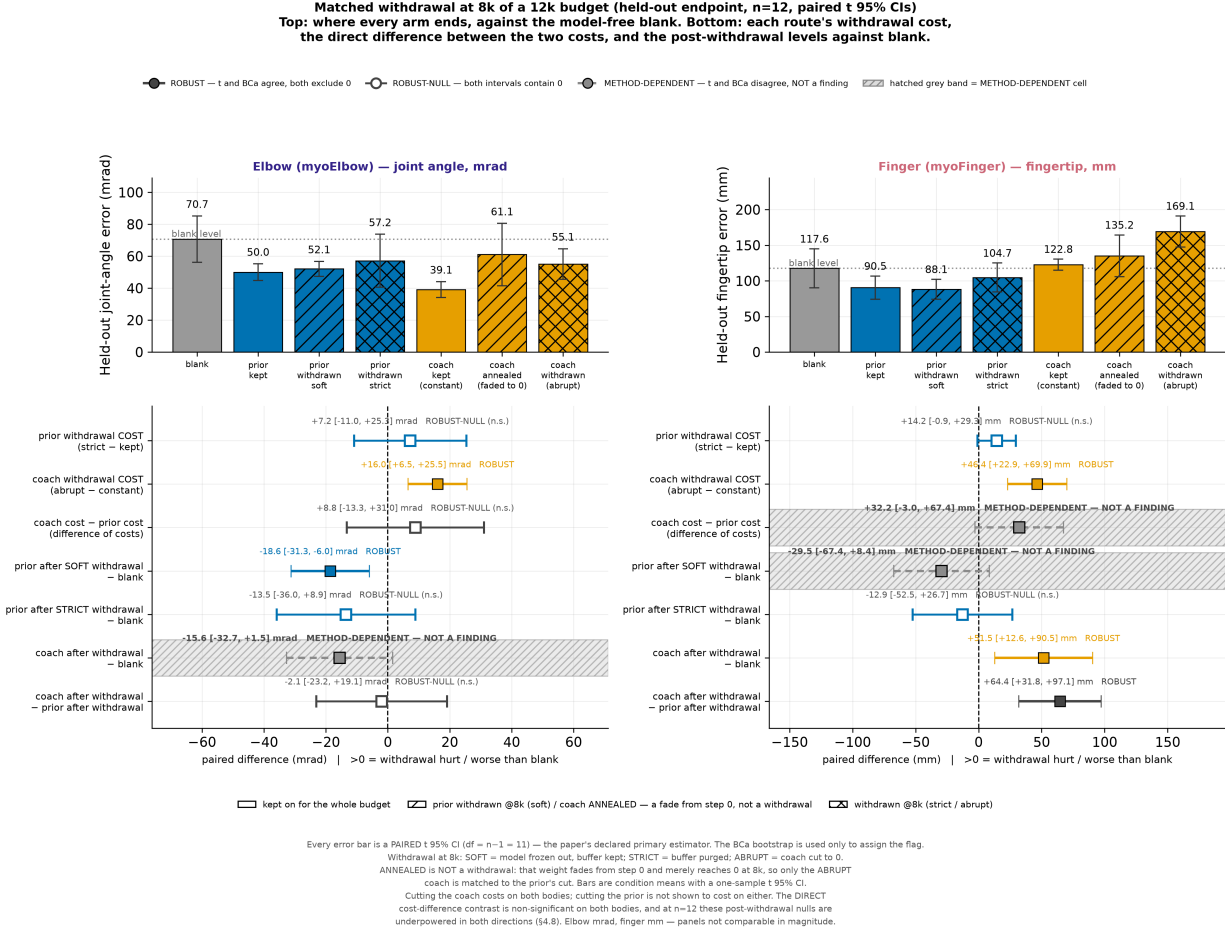


Figure 2: Withdrawal at 8k: per-route withdrawal costs, their direct difference, and post-withdrawal levels against model-free SAC, per body. Paired t 95% CIs, n=12.

window	elbow	finger
EARLY {2,4,6k}	-50.56 [-72.36, -28.76]	-11.72 [-19.29, -4.15]
MID {6,8,10k}	-22.46 [-33.81, -11.11]	-9.09 [-16.55, -1.62]
LATE {10,12k}	-11.83 [-19.52, -4.15]	+19.95 [+7.72, +32.18]
ENDPOINT {12k}	-10.89 [-19.26, -2.52]	+32.27 [+16.49, +48.05]

Table 6: coach(const) — prior by window, held-out. Negative means the coach is ahead.

The decay is the general phenomenon. The within-seed window $\times$ condition interaction [coach(const) — prior]@LATE — @EARLY, positive meaning the coach's lead shrinks, is +38.73 [+11.15, +66.31] on elbow and +31.67 [+15.21, +48.14] on finger. Those are raw units. The decay holds in raw units and under blank-normalisation on both bodies (elbow +0.2549 [+0.0430, +0.4669]); on myoElbow alone it does not survive the log-error reading (+0.3199 [-0.0070, +0.6467], $p = 0.0543$), because the elbow prior/coach error ratio only runs $\times 1.751$ (EARLY) $\rightarrow \times 1.272$ (LATE) while both arms' errors fall steeply. On finger all three scalings agree (log +0.2969 [+0.1226, +0.4712]; blank-normalised +0.2597 [+0.1344, +0.3850]) and what happens there is a sign change rather than a shrinkage, which no monotone rescaling can remove. What the two bodies differ in is whether the shrinking gap reaches zero inside the budget: on elbow it runs $-50.56 \rightarrow -22.46 \rightarrow -11.83 \rightarrow -10.89$, still on the coach's side of zero when the budget ends. On finger it

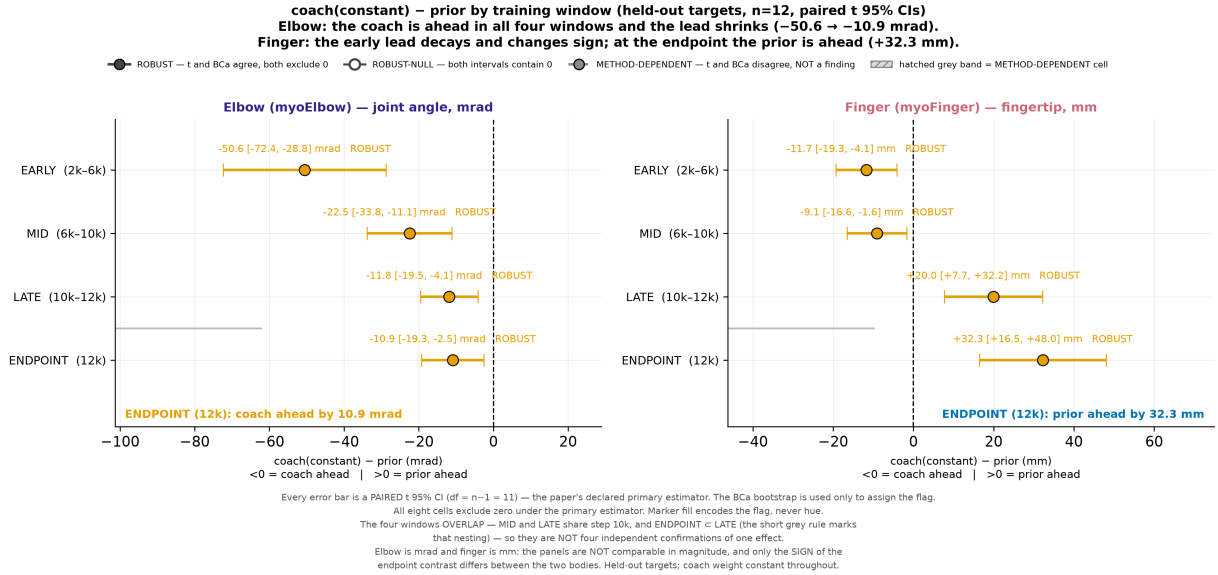


Figure 3: `coach(const)` – prior at EARLY/MID/LATE/ENDPOINT for each body. Negative = the coach has lower error. Error bars are paired t 95% CIs, n=12. All eight cells exclude zero under the primary estimator. MID overlaps EARLY and LATE and ENDPOINT ⊂ LATE, so the four points per body are not four independent confirmations (§3.6).

contrast	body	mean	paired t 95% CI
trained prior – randprior	elbow	−49.25	[−66.72, −31.78]
trained prior – randprior	finger	−40.81	[−57.24, −24.39]
randprior – blank	elbow	+23.75	[+1.72, +45.78]
randprior – blank	finger	+6.53	[−20.71, +33.76]

Table 7: The adversarial control for the forward-model route, scored on training targets at the endpoint, the only window and target set on which it was run.

runs $-11.72 \rightarrow -9.09 \rightarrow +19.95 \rightarrow +32.27$, changing sign. We establish a shared sign for the interaction and assert nothing about the shape of the path between the windows.

The reversal is not monotone and is carried by the last 2k: the coach leads at 4k, 6k and 8k ($-17.75, -16.03, -18.88$) and the 2k contrast contains zero, the 10k contrast is $+7.64$ with an interval containing zero, and we localize the crossing no more tightly than 8k–12k. The teacher ceiling of the teachers as captured is the leading account of where it falls; the finger prior falls 107.31 to 90.50 over exactly the last 2k that carries most of the $+32.27$, which a ceiling on the coach says nothing about. Appendix O gives the six-checkpoint table and that account in full, and Figure 3 plots the window contrast on both bodies.

4.6 The adversarial controls

The central objection to any frozen-model benefit is that *any* frozen model in a Dyna loop might regularize learning, regardless of whether it captures real body dynamics. We test it with an adversarial worst-case control: a random-initialized frozen world model in identical Dyna machinery (same k, `real_ratio`, `dyna_every`, `dyna_starts`), scored on training targets only in the analysis reported here (§6).

A random-initialized frozen model in the identical machinery is significantly worse than model-free RL on elbow, and the babble-trained model beats it on both bodies. The claim stops there. We do not claim from this control that learned body dynamics are the isolated active ingredient: a random network corrupts the analytic reward evaluated *on* its predicted states as well as the states themselves, so the contrast cannot

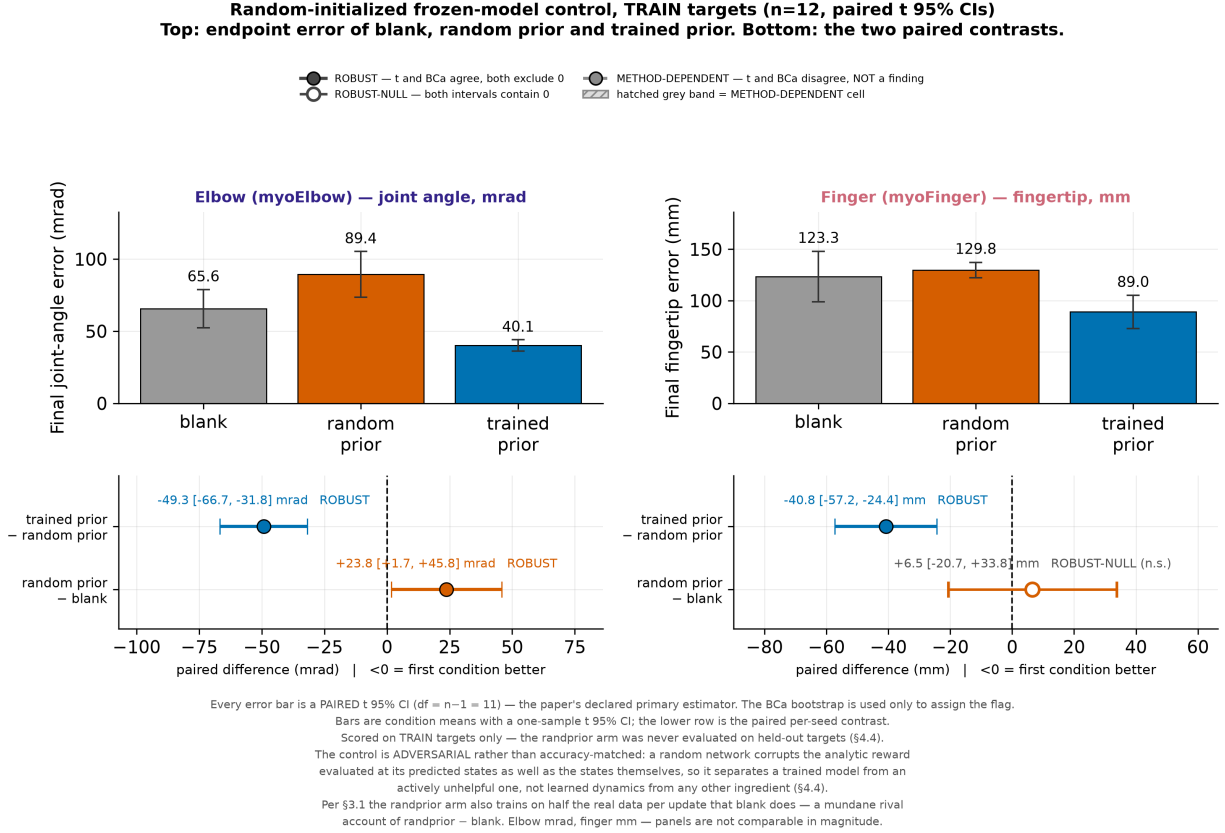


Figure 4: The random-init frozen-model control on train targets: **trained prior - randprior** and **randprior - blank** per body, paired t 95% CIs, n=12.

separate *learned dynamics* from *not-actively-adversarial dynamics*; the control is adversarial and is not accuracy-matched; and per §3.1 the **randprior** arm also trains on half the real data per update that **blank** does, which is a mundane rival explanation for **randprior - blank** being positive. Figure 4 plots the control.

4.7 The combination, and freezing

At the endpoint we cannot detect a difference between **priorcoach(const)** and **coach(const)** on either body, which per §6 is a failure to detect at n=12 and not a demonstration that combining lands on the coach's outcome. The combination is ahead of the coach alone in the elbow EARLY window ($-8.74 [-17.09, -0.38]$), so the prior is not inert in the presence of a maintained coach. A mixture-midpoint contrast rejects a 50/50 account on myoElbow at the endpoint and EARLY; Appendix D carries the eight mixture cells and the sixteen combination cells.

Frozen-ness is a design commitment carried by the title, so it is tested directly at n=12 on myoElbow. **prior - warm**, the identical babble checkpoint with **freeze=False**, is $+0.68 [-2.64, +4.01]$, excluding any effect beyond about $-2.6/+4.0$ mrad against the prior's own -25.50 mrad effect on that target set. Freezing neither helps nor hurts at a scale that could matter here. What that experiment does not resolve is babbling: the babble-pretrained arms separate from the online-from-scratch **colearn** arm on one axis only, seed-to-seed dispersion, where **colearn** is between about 1.7 and six times as dispersed and the two babble arms are indistinguishable from each other. What separates them is reproducibility, not accuracy, and what produces that dispersion is not identified: **colearn** differs from its babble-pretrained twin in five

implementation respects, not only in babble-pretraining. The comparison is post hoc and the arms are myoElbow train-target only. Appendix N gives the cells and the five differences.

4.8 Seed-to-seed dispersion

This analysis was not planned. It tests no part of the thesis and is not a result the paper argues from; it is reported because the withdrawal nulls invite the question. We report per-condition held-out endpoint SD across the 12 seeds and the ratio $SD(\text{blank})/SD(\text{condition})$ with percentile bootstrap 95% CIs, alongside the scale-free CV ratio, because reach error is bounded below by zero and SD tracks the mean. Appendix G carries both tables, Figure 5 and the full testing.

An aid still attached at 12k compresses seed-to-seed dispersion in three of the four kept-aid cells on the SD ratio and in two of the four on the CV ratio. After the two step-function withdrawals, the strict prior and the abrupt coach, dispersion is not distinguishable from the model-free baseline, a result underpowered in both directions, and this does not generalize to withdrawal as such: the elbow soft prior withdrawal is the tightest of the three withdrawn elbow arms. The two bodies disagree on whether the compression depends on what the model knows. On myoElbow an uninformative frozen model does not suppress dispersion while the trained prior does; on myoFinger the ordering inverts, and both directions are significant on the axis Appendix G declares. Both cells are post hoc, bodies are never pooled inside a contrast (§3.6), so no test across bodies is available and the disagreement is described but not established.

4.9 Competence referents and a long-budget probe

We fix two referents so "how good is this?" has an answer: a zero-activation no-op policy and a random-action policy, both scored under the identical protocol, and a condition is called *solved* when its error is an order of magnitude below the random-action referent and far below the no-op referent. Eight of the ten elbow conditions are solved in that sense; model-free **blank** and the random-teacher control **randcoach** are not. myoFinger is solved by no condition, so the finger endpoint result says which of two poor conditions is less poor. On myoFinger the no-op referent beats the random-action one by 71 mm, so inaction is a good policy there and thrashing an actively bad one, and the clean finger teacher sits within 3.2 mm of the no-op referent, which is why §5 disqualifies the finger cell from the guidance argument. The referents are unreplicated point estimates and an $n=3$ long-budget **blank-versus-prior** probe is reported alongside them; Appendix L carries both, with the re-drawn finger referent that moved 12.5%. The four unaided long-budget elbow runs that bear on amortization are in Appendix K (§5).

4.10 myoHand

Four MyoSuite bodies were instantiated and trained on. Two carry the analysis, the third is reported here and the fourth in Appendix F. On myoHand at $n=6$ over 60k steps we detect no improvement on either arm. One window cell reaches nominal significance and fails Holm over the unpruned five-cell window family, and over the three-cell family §3.6's exclusion criterion would derive; a 30-checkpoint grid returns the same verdict. We keep the body in the paper on the same terms as every other, precisely so that the exemption is not granted where it helps and imposed where it hurts. The body is uninformative about the forward-model route; Appendix L gives the cells.

4.11 Multiplicity

The uncorrected paired t interval is the primary reported statistic (§3.6). We also computed Holm–Bonferroni over six declared definitions of the correction family, because the reversal's survival depends on which definition is used. It survives four of the six, including the one §3.6's own exclusion criterion derives ($K = 44$), and fails the unpruned literal set ($K = 72$, or 88 once **randcoach** is entered); one of the four is the declared 28-cell set, which is significance-selected and which we report as such. We also report the mirror prune that deletes the reversal's own cell, so the enumeration is not chosen to favour it. Seven cells survive every definition in which they appear: both **trained prior** – **randprior** cells, both **coach(const)** – **randcoach** cells, the two finger **coach** – **blank** EARLY cells and the elbow **coach** – **prior** EARLY lead. Appendix

B carries the six enumerations, their K and their survivor counts; RECOMPUTED.md Appendices C–G carry the full K-sensitivity sweep.

5 Discussion

Take the guidance arm first. The guidance hypothesis (Salmoni et al., 1984) predicts that guidance aids acquisition-phase performance but not necessarily what is retained, and its two limbs need two measurements. §2 notes that its best-known test supports the retention limb more clearly than the acquisition limb. The acquisition limb is supported, on speed across both analysed bodies (coach – blank EARLY: elbow –59.0, finger –17.6) and on elbow endpoint accuracy (–31.6 against model-free, –10.9 against the prior). Those **coach – blank EARLY** cells measure a student under a weight- ≈ 1.0 behavioural-cloning pull toward a teacher that consumed $2.5\times$ the student’s entire budget in rewarded on-task steps, and find that student to have lower early error than an unaided one. That is closer to a pipeline sanity check than to a discovery, and we report it as one.

On the retention limb, removal significantly degrades the coach arm’s own endpoint accuracy on both bodies (+16.0, +46.4). We claim nothing about *what survives*. The finger cell is disqualified from the guidance argument, because the finger teacher as captured sits essentially at the no-op referent: 169.1 mm against 172.3, an unreplicated 3.2 mm that is not distinguishable from doing nothing. What that cell measures is the cost of removing a bad distillation target. That leaves one clean test, on elbow, coming out as a performance-versus-retention dissociation *within* the coach arm. The elbow teacher’s accuracy does transfer to the student while guidance is maintained (constant-coach student 39.13 mrad), and it is significantly degraded by removal (+16.00 mrad). Whether any of it survives is not resolved, since the elbow post-withdrawal cell against **blank** is significant under BCa only.

Turn to the forward-model arm. Consistent with internal forward-model theory (Wolpert et al., 1995; Kawato, 1999; Wolpert & Kawato, 1998) and dynamics adaptation (Shadmehr & Mussa-Ivaldi, 1994), the *learned* frozen forward model improves endpoint accuracy over model-free RL on elbow, on both target sets, at 12k steps, and the **randprior** control shows that an uninformative frozen model in the same machinery does not reproduce this on the train targets, the only set on which it carries a claim; §6 records unplanned held-out runs of the same arm.

Four qualifications follow from our own controls. First, the shape is unusual: on elbow the benefit is not detected EARLY (–8.47 [–31.82, +14.88], an interval that does not exclude an effect the size of the endpoint one), not detected in MID, and present LATE and at the endpoint, the reverse of the classical model-based story (Janner et al., 2019; Chua et al., 2018). That shape is three window verdicts, not a tested growth, and the interaction §3.6 requires for any growth claim is null on both bodies. Second, the arm operationalizes a forward model *plus* a known analytic reward evaluable at imagined states, so any biological reading must assume the nervous system has something like a state-evaluable cost at unvisited states, and the reversal is as much an oracle result as a forward-model one. Third, the elbow model’s own 19.3 mrad open-loop prediction error is essentially the size of the –20.7 mrad endpoint effect attributed to its fidelity. That figure still has no archived build transcript, but the checkpoint survives and reproduces it on re-measurement, and the off-distribution worry it carried now has a number: on the on-policy distribution the error is 24.9 mrad at five steps and 10.4 mrad at the horizon the Dyna loop uses (§3.3). Fourth, both endpoint accuracy results have an amortization reading the four unaided long-budget elbow runs do not exclude. Appendix K gives all four in full.

The two routes can now be compared directly. While present, the coach is the stronger aid at the myoElbow held-out endpoint and faster than the prior on both bodies, and that lead decays on both, crossing zero inside a 12k budget on myoFinger and not on myoElbow. The comparison is the contrast most exposed to the design asymmetries of §3.1 and §3.5: the coach arm trains on twice the real data per update while the prior arm consumes far more raw interaction and holds a reward oracle the coach does not. Every claim this paper makes about an aid beating model-free RL at the endpoint is an elbow claim, and the restriction to the endpoint is load-bearing because the EARLY window runs the other way on finger. Appendix K gives the cells.

That leaves the controls, and what the guidance control settled. The two routes are not equally well controlled: **randprior** ran throughout while **randcoach** was added last, and the gap mattered exactly where the guidance result looks strongest. The clean finger teacher as captured scores worse than a blank student and within 3.2 mm of the no-op referent, yet the finger coach still produces a large EARLY lead, so "distil the student toward a near-no-op action target on the body where inaction is worth 71 mm over thrashing" predicts an early gain of the observed sign independently of anything the teacher encodes. The control separates the two: on myoFinger EARLY a randomly-initialized teacher is itself a fixed non-flailing target and does not reproduce the coach's early gain, and on myoElbow it is significantly worse than no teacher at every window, the shape **randprior** shows for the forward-model route on the same body at the endpoint, the only window on which it was scored. On myoFinger **randcoach** is not distinguishable from model-free RL at EARLY or at the endpoint, and **randprior** is not distinguishable from it at the endpoint on train targets, the only window and target set on which it was scored; **randcoach** is significantly worse than model-free RL at MID (+12.86 [+8.12, +17.60]) and LATE (+26.44 [+11.86, +41.02]), the same shape it shows on myoElbow at every window (§4.2, §4.6).

One thing does constrain the rival directly. A movement index recoverable from the archived **eval_return** separates all eleven finger arms from the zero-activation referent by at least 29.7 mm on its own target set, the **randcoach** arm computed for this purpose, so no finger arm reproduces the no-op policy's movement signature. What it cannot do is measure how *much* an arm moves, which is what the rival actually turns on; the **randcoach** control settles that (Appendix K).

Two design points generalize beyond this study. An aid's measured benefit is a joint property of the aid *and* of the schedule under which it is withdrawn: our tapered coach faded from step 0 while the prior arm was bit-identical to its control until 8k, and comparing them as "withdrawn" arms would have measured two different manipulations. And a post-withdrawal null is a statement about dispersion as much as about means. The dispersion comparison that matters is against the *untreated* baseline and never against the treated arm's own earlier self.

6 Limitations

- The study is single-task reach and simulation-only, no hardware, no sim-to-real, on deterministic, fully-observed bodies, conditions that *favour* a frozen forward model.
- Four MyoSuite bodies were instantiated and trained on, and two are analysed. myoElbow is 1-DoF, so its 20 held-out targets interpolate between the 8 training targets, and its generalization test is much weaker than the finger's. On myoHand (§4.10) we detect no improvement on either arm within 60k steps (+3.9 [−7.1, +14.9] and −3.5 [−13.8, +6.8]), and the body is uninformative about either route. myoArm (Appendix F) is a single truncated **blank** run under MyoSuite's own reward and carries no claim.
- $n=12$ seeds for all main results and $n=6$ on myoHand. Every non-significant cell should be read as a failure to detect at that power and never as evidence of equivalence, except where an interval is narrow enough to exclude an effect of the size at issue and the comparison is made explicitly, as for **prior** – **warm** in §4.7.
- Neither withdrawal is a clean single-factor manipulation. Strict withdrawal removes imagination *and* doubles the real-data gradient signal per update, because purging the buffer permanently satisfies the update rule's guard. Soft withdrawal keeps the 64 + 64 mix but on a stale buffer. The coach's abrupt withdrawal is cleaner but not single-factor either: it removes the teacher's information *and* deletes a weight- ≈ 1.0 auxiliary term from the loss in one step (§4.4), and Appendix O attributes the finger cost to that optimization shock, not to loss of teacher content. The control separating them was not run. Every withdrawal contrast in §4.4 carries this (§3.4), and the confound pushes the prior's removal cost toward zero.
- The Dyna arms train on 64 real + 64 imagined transitions per update while the model-free arms train on 128 real (**real_ratio** = 0.5), so equal update-to-data is not equal real-data gradient signal, and the matched-replay control was not run.

- The prior arm additionally receives the environment’s analytic reward evaluated at imagined states, task information no model-free arm has (§3.5).
- Contrasts were selected sequentially during the study and were not specified in advance, multiplicity is reported but not pre-registered, and the endpoint is a fixed 12k-step budget, not a verified asymptote.

The remaining eight limitations, none of which is load-bearing for a claim in §4 or §5, are in Appendix H: the myoHand rival reading, the single frozen checkpoint per body, the seven conditions without an archived launcher, the single teacher per body, the single target-set draw, the absent hyperparameter search, the unlogged behavioural variables, and the lateness of the guidance control together with its missing swap-at-8k half.

Claims withdrawn during the study. Five claims were withdrawn, in every case after the control or the direct paired contrast that tested them was run: a double dissociation between the two routes, a back-loading of the prior’s advantage, a withdrawal asymmetry between routes, a decay of the prior’s contribution to the combination, and a dispersion "inversion" that is a null (§4.8). Separately, an interval-estimator error, a normal-approximation multiplier ($z = 1.96$) where a t multiplier is required at $n=12$, was corrected by recomputing all 132 contrasts under the policy of §3.6, a correction that could only remove significance and never create it. The itemised record is in `RECOMPUTED.md`, and any completeness statement about this repository is scoped there to *no unreported artefact remains that writes a file into this tree*, with named exclusions: thirty-two further 12k-step runs sit under `actors_withdraw/`, `actors_randcoach/`, `actors_finger/` and `actors_video/`, all trained after this analysis was complete solely to render the comparison videos. Twenty-seven re-execute reported conditions, bit-identically where we have checked. The other five score `randprior` held-out, which the study itself never reports: one on myoElbow at seed 0 (`actors_video/randprior_actor.pt`, evaluation output not redirected to a file and unrecoverable) and four on myoFinger at four seeds: `randprior - blank = +37.30 [-31.00, +105.60]` and `randprior - prior = +63.70 [+11.26, +116.14]`. They were not planned as an arm, the finger four are $n=4$, and they carry no claim. Appendix H gives the directory counts and the bit-identity check.

7 Conclusion

Guidance and a body model are not interchangeable. Under one SAC backbone, $n=12$ matched seeds and paired t intervals, guidance buys the early speed: a distilled coach held at constant weight leads a frozen babble-learned body model early on both analysed bodies, and that lead decays over the budget even while the coach stays attached. Cutting the coach is separately costly on both, though only the elbow cost reads as a loss of what the teacher knew (§5). What the model buys arrives later, and inside a 12k-step budget the decay crosses zero on myoFinger and not on myoElbow. An actively wrong aid does not substitute for a right one: a babble-trained frozen model beats a random-initialized one on both bodies, on train targets, and a randomly-initialized teacher in the identical loss is worse than no teacher on myoElbow and does not reproduce the coach’s early gain on myoFinger, which closes the rival account §5 could not previously separate.

Four qualifications belong in the same breath. The prior arm evaluates the environment’s analytic reward at its imagined states, so the endpoint reversal is scoped to a forward model *plus a known reward function* and is as much an oracle result as a forward-model one (§3.5). Cutting the prior is not shown to be costly on either body and the direct removal-cost difference between the routes is non-significant on both, a null with a named candidate generator (§3.4). The decay holds in raw units and under blank-normalisation on both bodies and, on myoElbow alone, not under the log-error reading (§4.5). And the reversal’s status under multiplicity correction is family-dependent (§4.11). Freezing itself was tested at $n=12$ on myoElbow and is inconsequential at this scale (§4.7).

Next steps. The cheapest is to log at least one behavioural variable at every evaluation checkpoint and to resample the target sets per seed. Then, in rough order of cost, the swap half of the `randcoach` control at the 8k withdrawal point, a matched-replay Dyna control, and `blank` to 30k at $n=12$, which would settle the amortization reading of §5. Appendix J lists the remainder.

#	contrast	body	mean	paired t	95% CI	BCa 95% CI
1	prior – blank (held-out, ENDPOINT)	finger	−27.10	[−66.32, +12.12]	[−80.47, −3.85]	
2	prior – blank (train, ENDPOINT)	finger	−34.29	[−70.93, +2.35]	[−72.79, −8.91]	
3	prior – blank (held-out, EARLY)	finger	−5.88	[−16.15, +4.38]	[−20.25, −0.11]	
4	coach cost – prior cost (ENDPOINT)	finger	+32.18	[−3.02, +67.39]	[+6.59, +68.56]	
5	coach abrupt-withdrawn – blank (ENDPOINT)	elbow	−15.60	[−32.75, +1.54]	[−30.09, −0.87]	
6	prior soft-withdrawn – blank (ENDPOINT)	finger	−29.49	[−67.41, +8.43]	[−78.26, −6.21]	
7	prior soft-withdrawn – prior strict (ENDPOINT)	finger	−16.58	[−33.65, +0.49]	[−30.91, −1.86]	
8	[priorcoach(const) – coach(const)] @EARLY – @ENDPOINT	elbow	−8.10	[−18.06, +1.87]	[−17.30, −0.22]	
9	coach clean – coach contaminated (train, ENDPOINT)	finger	+37.71	[−1.62, +77.03]	[+4.90, +72.21]	
10	priorcoach(const) – 0.5·[prior + coach(const)] (ENDPOINT)	finger	+9.50	[−1.02, +20.01]	[+1.53, +19.69]	
11	priorcoach(anneal) – coach(anneal) (held-out, EARLY)	elbow	−10.90	[−22.43, +0.64]	[−20.48, −0.91]	
12	[coach(anneal) – blank] @ENDPOINT – @EARLY	elbow	+38.61	[−6.87, +84.08]	[+2.60, +79.94]	
13	prior – blank (held-out, MID)	elbow	−12.25	[−25.34, +0.84]	[−22.91, −0.63]	
14	coach(anneal) – prior (held-out, LATE)	elbow	+21.52	[−1.06, +44.10]	[+5.97, +46.17]	

Table 8: The fourteen contrasts on which the paired t and the BCa bootstrap disagree. Under §3.6 they carry no claim anywhere in this paper.

A BCa-only contrasts

These 14 of the 134 contrasts in the `RECOMPUTED.md` register described in Appendix I are the ones on which the primary (paired t) and the BCa bootstrap intervals disagree. Under the policy of §3.6 they carry no claim anywhere in this paper. In every one the direction of disagreement is the same, BCa excludes zero and t does not, so a reader who preferred BCa as primary would reach fourteen further positive verdicts. A t-non-significant flag here is not a claim that the effect is absent.

B Multiplicity: the six family definitions

The uncorrected paired t interval is the primary reported statistic (§3.6). We also computed Holm–Bonferroni at $\alpha = 0.05$, and whether the finger endpoint reversal of §4.5 survives depends on how the contrast family is defined. Over the 28 declared claim-bearing contrasts, 16 survive and the reversal is rank 7. Those 28 are, however, a 28-cell subset of the 38 t-significant cells §3.6’s enumeration rule generates, and correcting inside a set selected on significance understates the correction. Enumerated literally, one contrast per (condition pair $\times$ window) among the directly compared conditions, both bodies, plus the two withdrawal-cost differences and the two decay interactions, the rule yields $K = 72$, of which 38 are t-significant, 12 survive, and the step-down stops on the reversal itself ($p = 0.00090$ against a 0.00083 threshold), taking the finger decay interaction, both coach withdrawal costs and every §4.3 endpoint accuracy cell with it. But that literal set violates §3.6’s own exclusion criterion, which forbids cells that sub-divide a window already in the family: $\text{ENDPOINT} \subset \text{LATE}$ and MID overlaps both, and the literal 72 admits all of them. Pruning the duplicated LATE cells gives $K = 58$ (reversal rank 12, survives) and pruning MID as well gives $K = 44$, what the criterion actually derives, on which the reversal is rank 9 against a 0.00139 threshold and survives with room, as do both finger withdrawal cells and the finger decay interaction. A by-question split into route-comparison ($K=26$), versus-model-free ($K=36$) and removal ($K=10$) families also keeps it (rank 5 of 26), and rescues both coach withdrawal costs and two of the four §4.3 endpoint accuracy cells. So the reversal survives four of the six family definitions we report, including the one our own rule derives, and fails the unpruned literal set, which is itself not selection-free. A sixth family we computed cuts the other way: pruning ENDPOINT instead of LATE for the same seven window-spanning pairs also gives $K = 58$, but it removes the title’s own cell from the family, so a reader who prefers that prune direction gets no reversal cell to correct and no surviving one (`RECOMPUTED.md` F1d). Which member of an $\text{ENDPOINT} \subset \text{LATE}$ pair is redundant is a choice, and it determines whether the title claim has a cell at all. We report both directions. What holds under every definition in which it appears, and at every widening of the enumeration we checked, is seven cells: both `trained prior – randprior` cells, both `coach(const) – randcoach` cells, the two *finger coach – blank* EARLY cells, and the elbow *coach – prior* EARLY lead. Four of the seven are adversarial controls and two are the sanity checks of §5. One is a route comparison. The literal enumeration

family	K	t-significant	Holm survivors	verdict on the reversal
declared (the claim-bearing set: the t-significant sub-set, minus ten)	28	28	16	survives — rank 7, threshold 0.00227
literal (the enumeration, unpruned)	72	38	12	fails — the stopping cell, threshold 0.00083
LATE pruned (partial prune of ENDPOINT $\subset$ LATE)	58	31	12	survives — rank 12, threshold 0.00106
MID + LATE pruned (what §3.6’s criterion derives)	44	25	12	survives — rank 9, threshold 0.00139
by-question split, route-comparison family	26	19	8	survives — rank 5, threshold 0.00227

Table 9: The six definitions of the correction family, with each one’s K, its t-significant count, its Holm survivor count and its verdict on the myoFinger reversal.

window	blank (mm)	prior (mm)	prior – blank	paired t	95% CI	t p	exact Wilcoxon p	seeds on the sign
EARLY {2,4,6k}	106.29	105.63	−0.66	[−7.40, +6.08]	0.812		1.000	3/6
MID {6,8,10k}	108.21	110.08	+1.87	[−6.81, +10.55]	0.604		0.688	3/6
LATE {10,12k}	105.30	115.71	+10.42	[−5.39, +26.22]	0.151		0.219	4/6
12k	106.12	118.61	+12.49	[+0.99, +23.99]	0.038		0.03125	6/6
60k (this run’s endpoint)	106.34	99.55	−6.79	[−22.45, +8.86]	0.315		0.563	4/6

Table 10: myoHand, paired over the six shared seeds. Positive means the prior is worse. $n=6$, so the multiplier is $t(.975, 5) = 2.5706$.

and a summary table follow, carrying one row for each of the five families that contain the reversal’s cell, with each family’s K, its t-significant count, its Holm survivor count and its verdict on the reversal.

The literal enumeration of §3.6’s rule is: seven directly-compared condition pairs $\times$ four windows $\times$ two bodies (56 cells: **prior** – **blank** held-out and train, **coach(const)** – **blank**, **coach(anneal)** – **blank**, **coach(const)** – **prior**, **priorcoach(const)** – **prior**, **priorcoach(const)** – **coach(const)**); two **randprior** pairs at ENDPOINT $\times$ two bodies (4); four withdrawal pairs at ENDPOINT $\times$ two bodies (8); the withdrawal-cost difference (2); and the decay interaction (2), $K = 72$. That enumeration predates the **randcoach** control (§3.2), and entering its two pairs on the same footing as the other directly-compared pairs adds $2 \times 4 \times 2 = 16$ cells: $K = 88$ literal, and 52 under the MID+LATE prune §3.6’s criterion derives. Both verdicts are unchanged, the reversal still fails the literal set, and still survives the derived one at rank 12 against $0.05/41 = 0.00122$, and both **coach(const)** – **randcoach** EARLY cells ($t = 7.81$ and 7.15 , $df = 11$) enter above the reversal in every family we computed, so they survive wherever anything does. The table below is printed at the pre-**randcoach** K. The recomputed values are the ones in this paragraph. Five of the six definitions we report are tabulated below with their verdict on **coach(const)** – **prior** ENDPOINT finger ($p = 0.00090$). The sixth, the mirror prune that removes that cell from the family altogether, is described in §4.11 and **RECOMPUTED.md** F1d and has no row in Table 9 because the cell it would be scored in does not exist.

The other two by-question families are versus-model-free ($K = 36$, 12 survive) and removal ($K = 10$, 3 survive); $26 + 36 + 10 = 72$, and 23 cells survive across the three against 12 under the single literal family. Holm is valid under arbitrary dependence, so including correlated cells is conservative and not erroneous, and the literal-72 verdict is correct *about that family*. It is not, however, a selection-free enumeration: it admits **coach abrupt** – **blank** for both bodies while admitting **prior strict-withdrawn** – **blank** for neither, dropping the non-significant member of a matched pair. Widening it matters for the survivor count but not for the stopping cell: rank 12 of the literal family clears its threshold by only $1.12\times$, so eight further less-significant cells take the both-family core from six to five. Cell-by-cell tables for all five families, the K-sensitivity sweep and the three by-question families are in **RECOMPUTED.md** Appendices C–G.

C myoHand

Paired over the 6 shared seeds, **prior** – **blank** on train targets, positive = the prior is worse. $N = 6$, so the multiplier is $t(.975, 5) = 2.5706$.

Levels at 2k $\rightarrow$ 12k $\rightarrow$ 60k: blank 102.5 $\rightarrow$ 106.1 $\rightarrow$ 106.3; prior 103.1 $\rightarrow$ 118.6 $\rightarrow$ 99.6. Within arm across the budget: blank 60k $-$ 2k = +3.88 [−7.10, +14.86], prior −3.51 [−13.81, +6.80]. On mean(last 5 checkpoints) $-$ mean(first 5), blank −0.28 [−6.28, +5.71] and prior −3.82 [−12.48, +4.83].

Holm over these five window cells, the unpruned window family for this body; the two analysed bodies are corrected under the MID+LATE-pruned $K = 44$ family of §4.11: rank 1 is 12k at $p = 0.03832$ against $0.05/5 = 0.01000$, it fails, and the step-down stops there, so no cell survives. The per-checkpoint grid, computed as well and reported for disclosure and not used: two of the thirty archived checkpoints reach $p < 0.05$ (12k at 0.0383, 38k at 0.0359) against a chance expectation of 1.5, and Holm over that grid also stops at rank 1.

D Additional contrasts

Teacher decontamination (clean $-$ contaminated; positive = decontamination made the student worse). On elbow all eight cells are non-significant, including held-out ENDPOINT (+2.34 [−17.10, +21.77]) and EARLY (+1.85 [−5.20, +8.90]). On finger decontamination significantly deflated the early advantage in all four EARLY cells (coach +15.21 [+9.43, +20.98] held-out and +23.76 [+16.93, +30.59] train; priorcoach +13.72 and +27.29) while leaving the held-out endpoint non-significant (+18.48 [−21.11, +58.07]). The finger anti-synergy survives decontamination: priorcoach(anneal) $-$ prior at ENDPOINT is +23.99 [+5.46, +42.52] held-out and +26.31 [+9.46, +43.17] train.

Coach schedule. coach(const) $-$ coach(anneal), held-out: ENDPOINT elbow −21.94 [−40.88, −3.01], finger −12.38 [−39.89, +15.13]; EARLY elbow −10.76 [−15.09, −6.42], finger −0.21 [−4.28, +3.86]; MID elbow −36.81 [−51.81, −21.82], finger −14.87 [−27.51, −2.23]; LATE elbow −33.35 [−54.23, −12.47], finger −15.42 [−41.16, +10.32].

Combination, full window table. priorcoach(const) $-$ prior: EARLY elbow −59.30 [−80.25, −38.34], finger −16.54 [−20.62, −12.46]; MID elbow −24.58 [−33.48, −15.68], finger −9.51 [−16.24, −2.78]; LATE elbow −11.43 [−18.43, −4.44], finger +16.23 [+3.83, +28.64]; ENDPOINT elbow −11.53 [−17.80, −5.27], finger +25.63 [+8.88, +42.38]. priorcoach(const) $-$ coach(const): EARLY elbow −8.74 [−17.09, −0.38], finger −4.82 [−10.18, +0.54]; MID elbow −2.12, finger −0.42; LATE elbow +0.40, finger −3.72; ENDPOINT elbow −0.64 [−4.05, +2.77], finger −6.64 [−14.71, +1.43].

Mixture midpoint. priorcoach $-$ 0.5·[prior + coach], where a deviation from zero rejects a 50/50 mixture account: constant schedule, ENDPOINT elbow −6.09 [−8.90, −3.27], finger +9.50 [−1.02, +20.01]. EARLY elbow −34.02 [−45.67, −22.36], finger −10.68 [−13.57, −7.79]. Annealed schedule, EARLY elbow −30.80 [−43.39, −18.21], finger −11.17 [−14.58, −7.76]. Both annealed ENDPOINT cells non-significant.

Tapered-schedule route contrast. coach(anneal) $-$ prior, held-out: EARLY elbow −39.80 [−61.37, −18.24], finger −11.51 [−18.31, −4.70]; MID elbow +14.35 [−0.83, +29.53], finger +5.78; LATE elbow +21.52 [−1.06, +44.10], finger +35.38 [+6.76, +64.00]; ENDPOINT elbow +11.05 [−8.89, +30.99], finger +44.65 [+12.12, +77.18]. Per §3.2 these cells measure a coach whose weight has been falling since step 0 and support no claim.

prior $-$ blank in the intermediate windows. MID elbow −12.25 [−25.34, +0.84], finger −1.32. LATE elbow −21.92 [−32.85, −10.98], finger −12.56 [−37.94, +12.81]. ENDPOINT $\subset$ LATE, so the elbow LATE cell is not independent confirmation of §4.3; coach-blank window interactions (positive = the gap in the aid’s favour is smaller later): [coach(const) $-$ blank]@LATE $-$ @EARLY elbow +25.28 [−9.34, +59.90], finger +24.99 [+6.33, +43.65]; [coach(anneal) $-$ blank]@LATE $-$ @EARLY elbow +47.88 [+3.26, +92.50], finger +40.20 [+11.06, +69.34].

E Additional figures

The figure set is in `wm_prior/figs/`. Every contrast error bar is the paired t 95% CI ($df = 11$), except Figure 5’s dispersion bars, which are percentile bootstrap intervals (§4.8); per-condition bars and learning-curve bands carry the corresponding one-sample t 95% CI; the BCa bootstrap is used only to assign flags. Elbow panels are mrad and finger panels are mm, no panel compares them numerically, and

`make_figs.py` verifies every plotted contrast against the recomputed table, value, interval and flag, before any image is written, except the four train-target `coach(anneal)` – `blank` cells, which are not in that table and are computed with the same estimators and flagged as such. Beyond the five figures in the body: `fig1_2x2_endpoint.png` endpoint bars, train and held-out, both schedules; `fig2_paired_contrast.png` coach – prior at ENDPOINT and EARLY under both schedules; `fig3_guidance_coachVblank.png` coach – blank across target sets and schedules; `fig4_heldout_vs_train.png` prior – blank on both target sets; `fig6_timecourse.png` coach – prior by window under both schedules; `fig8_plateau.png` the $n=3$ long-budget runs, individual seeds shown with no interval drawn; `fig9_mixture.png` the annealed-schedule mixture test; `fig11_coach_dominance.png` the combination against the prior alone, the coach alone and the 50/50 midpoint under the constant schedule. These appendix figures carry the per-cell Holm annotations that the body figures omit.

F myoArm, the fourth body

`gonogo_myarm.py` calls `gym.make("myoArmReachRandom-v0")` and trains this study’s SAC on it (`myoArm: obs 80 act 34 muscles 34`). `gonogo_myarm.log` records greedy fingertip–target distances falling from an initial ≈ 0.98 m to 0.417, 0.352, 0.564, 0.313 and 0.281 m at 2k–10k steps, at which point the run was killed against a declared budget of 40k. It is a single seed, `blank` only, at `hidden=256` instead of the study’s 128, under MyoSuite’s native reward rather than this paper’s reach reward, with no held-out target set, so it supports no contrast, no interval and no claim, exactly as this paper says of `myoHand`. It is reported because the body count belongs in the paper and because the run writes `stdout` only and creates no results directory, which is why the completeness register that enumerates `results*` directories could not have found it (`RECOMPUTED.md` H4, H5).

G Seed-to-seed dispersion, in full

A separate, declared apparatus governs the post-hoc dispersion analysis of §4.8. A dispersion is not a paired difference, so the paired t interval above cannot be applied to it and the two estimators above are not what §4.8 uses. §4.8 reports per-condition endpoint SD and CV together with the ratios $SD(\text{blank})/SD(\text{condition})$ and $CV(\text{blank})/CV(\text{condition})$, each with a percentile bootstrap 95% CI (20 000 resamples). Where it compares two dispersions instead of listing them side by side, it uses two further tests, declared here. The first is a Pitman–Morgan paired variance-ratio test over the seeds the two conditions share. The second is a paired bootstrap of the ratio-of-ratios (20 000 resamples), whose interval is read against 1. "Significantly tighter" in §4.8 means exactly this, a ratio-of-ratios interval excluding 1 with an agreeing Pitman–Morgan p , and nothing in §4.8 licenses a claim under any other estimator. These cells sit outside every multiplicity family in this paper for the same reason the ratios do: they are not paired-difference contrasts and carry no paired- t p -value the Holm procedure could take, and their exclusion is not presented as evidence that they would survive one. Everything in §4.8 is post hoc.

The subsection this supports is §4.8. Everything it summarises is here.

This analysis was not planned. It was prompted by a question about what the withdrawal nulls mean, it tests no part of the thesis, and it is reported as a secondary observation and is not a result the paper argues from. We report per-condition held-out endpoint SD across the 12 seeds and the ratio $SD(\text{blank})/SD(\text{condition})$, with percentile bootstrap 95% CIs (20 000 resamples). A ratio above 1 means the condition is *less* dispersed across seeds than model-free SAC. Because reach error is bounded below by zero, SD tracks the mean, so the scale-free CV ratio is reported alongside every SD ratio.

Table 12 repeats the measures on the 8 training targets, against the matched train-target `blank`.

An aid still attached at 12k compresses seed-to-seed dispersion in three of the four kept-aid cells on the SD ratio and in two of the four on the scale-free CV ratio. On SD: elbow prior 2.78, elbow coach 2.93 and finger coach 3.53 exclude 1, and the fourth cell, finger `prior` kept at 1.68 [0.54, 3.51], does not. On CV: elbow prior 1.97 and finger coach 3.69 exclude 1, and elbow coach falls to 1.62 [0.87, 3.73], joining the finger prior’s 1.29. The two combined `priorcoach(const)` arms carry the study’s numerically largest ratios (4.77 [2.44,

condition	body	mean	SD	CV	SD ratio	95% CI	CV ratio	95% CI
blank (reference)	elbow	70.73	22.78	0.322	1.00	—	1.00	—
prior kept	elbow	50.02	8.18	0.163	2.78	[1.48, 4.63]	1.97	[1.07, 3.28]
prior soft-withdrawn	elbow	52.11	7.34	0.141	3.10	[1.61, 5.46]	2.28	[1.27, 3.80]
prior strict-withdrawn	elbow	57.19	26.05	0.456	0.87	[0.45, 1.58]	0.71	[0.37, 1.28]
coach constant	elbow	39.13	7.77	0.199	2.93	[1.44, 7.25]	1.62	[0.87, 3.73]
coach annealed	elbow	61.07	30.66	0.502	0.74	[0.37, 1.47]	0.64	[0.35, 1.12]
coach abrupt-withdrawn	elbow	55.13	15.06	0.273	1.51	[0.78, 2.78]	1.18	[0.63, 2.12]
prior + constant coach	elbow	38.49	4.77	0.124	4.77	[2.44, 8.96]	2.60	[1.41, 4.68]
prior + annealed coach	elbow	57.72	13.98	0.242	1.63	[0.85, 2.92]	1.33	[0.72, 2.31]
blank (reference)	finger	117.60	43.03	0.366	1.00	—	1.00	—
prior kept	finger	90.50	25.57	0.283	1.68	[0.54, 3.51]	1.29	[0.42, 2.59]
prior soft-withdrawn	finger	88.12	22.14	0.251	1.94	[0.63, 3.74]	1.46	[0.49, 2.65]
prior strict-withdrawn	finger	104.69	31.98	0.305	1.35	[0.44, 2.68]	1.20	[0.40, 2.40]
coach constant	finger	122.77	12.18	0.099	3.53	[1.16, 6.67]	3.69	[1.28, 6.60]
coach annealed	finger	135.15	46.10	0.341	0.93	[0.30, 2.83]	1.07	[0.38, 2.64]
coach abrupt-withdrawn	finger	169.14	34.52	0.204	1.25	[0.41, 2.69]	1.79	[0.64, 3.39]
prior + constant coach	finger	116.13	9.19	0.079	4.68	[1.53, 8.78]	4.63	[1.62, 8.04]
prior + annealed coach	finger	114.49	17.84	0.156	2.41	[0.78, 4.80]	2.35	[0.80, 4.52]

Table 11: Seed-to-seed dispersion on held-out targets: per-condition SD and CV, and each one’s ratio against model-free SAC with a percentile bootstrap interval.

condition	body	mean	SD	CV	SD ratio	95% CI	CV ratio	95% CI
blank (train reference)	elbow	65.65	20.95	0.319	1.00	—	1.00	—
prior kept (train)	elbow	40.15	6.27	0.156	3.34	[2.12, 5.45]	2.04	[1.26, 3.54]
randprior (train)	elbow	89.40	25.08	0.281	0.84	[0.53, 1.27]	1.14	[0.70, 1.87]
blank (train reference)	finger	123.25	38.72	0.314	1.00	—	1.00	—
prior kept (train)	finger	88.97	25.19	0.283	1.54	[0.78, 2.84]	1.11	[0.55, 2.09]
randprior (train)	finger	129.78	11.65	0.090	3.33	[1.66, 7.59]	3.50	[1.81, 7.66]

Table 12: The same dispersion measures on the eight training targets, against the matched train-target blank.

8.96] elbow, 4.68 [1.53, 8.78] finger), though, tested on the axis declared at the head of this appendix, neither is distinguishable from the next cell on its body (elbow against **prior soft-withdrawn**, ratio-of-ratios 1.54 [0.83, 2.80], Pitman–Morgan $p = 0.190$; finger against **coach constant**, 1.33 [0.87, 2.04], $p = 0.362$), so the ranking is reported, not claimed. After the two step-function withdrawals, the strict prior and the abrupt coach, dispersion is not distinguishable from the model-free baseline (elbow strict prior 0.87, elbow abrupt coach 1.51), a result that is underpowered in both directions. This does not generalize to withdrawal as such: the elbow soft prior withdrawal, which differs from the strict one only in that the imagined-transition buffer is not purged, is at 3.10 [1.61, 5.46] the tightest of the three withdrawn elbow arms. Two SD point estimates are not a comparison, so that row is tested on the same axis as the rest of this subsection, over the twelve shared held-out seeds: the soft arm is significantly tighter than both step-function withdrawals, against the strict prior withdrawal, ratio-of-ratios 3.55 [2.27, 6.11], Pitman–Morgan $p = 0.00042$; against the abrupt coach withdrawal, 2.05 [1.19, 3.68], $p = 0.03204$, and it is not distinguishable from the kept arm it was withdrawn from: 3.10 against the kept arm’s 2.78 is a ratio-of-ratios of 1.114 [0.680, 1.896], an interval including 1, with Pitman–Morgan $p = 0.70198$, and the finger twin is 1.155 [0.809, 1.546], $p = 0.51932$. Read as two point estimates, the 3.10-versus-2.78 gap looks like an inversion of the expected direction. Tested on the axis this appendix declares for dispersion comparisons, it is not distinguishable from 1 (1.114 [0.680, 1.896], $p = 0.70$): the inversion is not established, and at this power neither is its absence.

The dispersion effect is not attributable to what the model knows, on finger. The adversarial control answers the objection’s twin, that any frozen model in a Dyna loop might contract spread regardless of content. On elbow the uninformative model does not suppress dispersion (0.84 [0.53, 1.27]) while the trained prior on

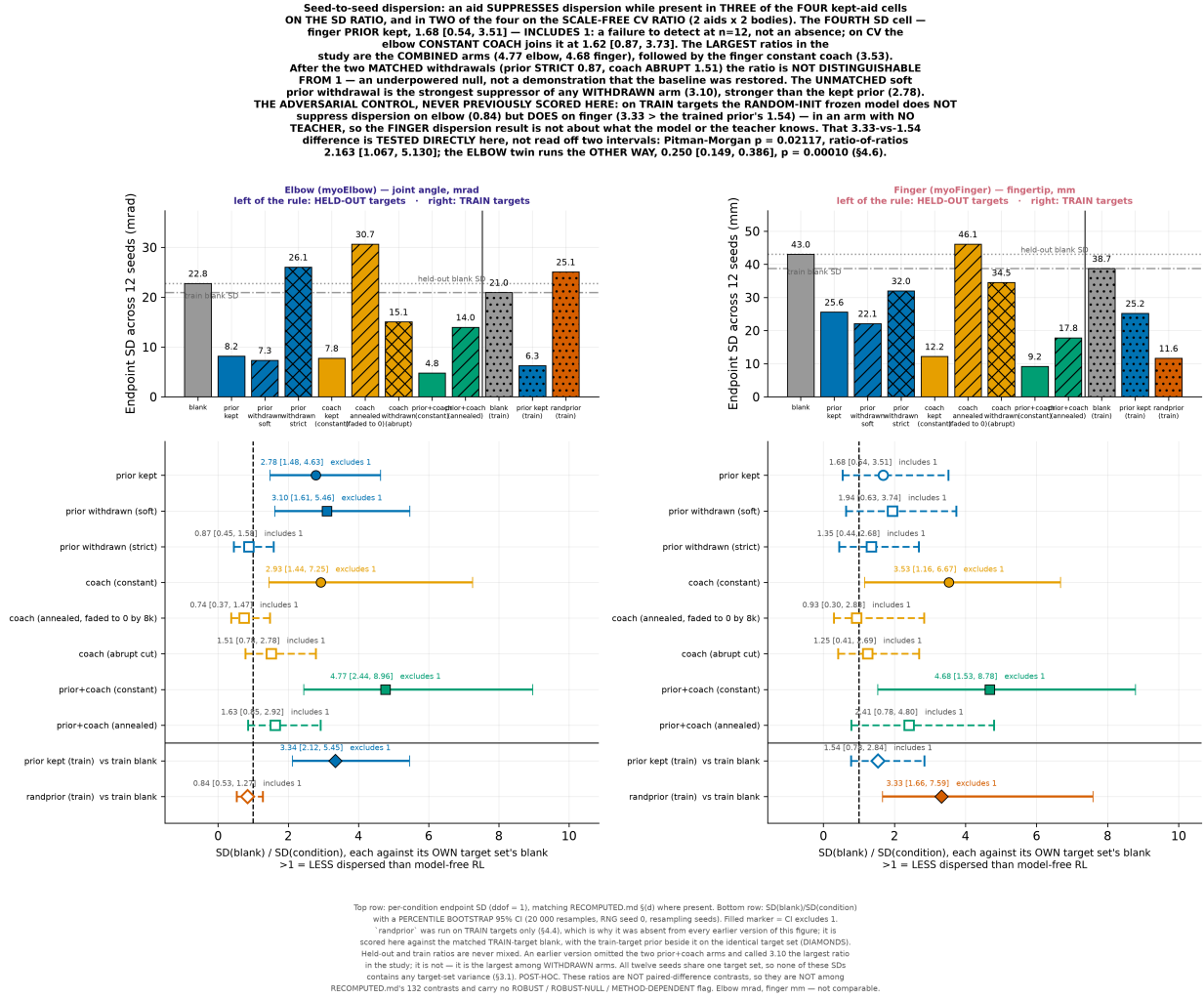


Figure 5: Per-condition endpoint SD across 12 seeds and the SD(blank)/SD(condition) ratios with percentile bootstrap 95% CIs, for the nine held-out conditions tabulated here per body (randcoach, added late, is in §4.1 but not this table), plus the three train-target rows scored against the matched train-target blank. Held-out and train ratios are never mixed.

the same targets is at 3.34. On finger the ordering inverts: randprior is at 3.33 [1.66, 7.59] against the trained prior's 1.54 [0.78, 2.84] on the identical target set, in an arm with no teacher at all. Because two separate interval verdicts do not constitute a comparison, the difference is tested directly over the twelve shared train-target seeds: a Pitman–Morgan paired variance-ratio test gives finger $p = 0.02117$, and a paired bootstrap of the ratio-of-ratios gives 2.163 [1.067, 5.130], excluding 1. The elbow twin is significant in the opposite direction (0.250 [0.149, 0.386], $p = 0.00010$). So the two bodies' point verdicts run in opposite directions on content-specificity. Bodies are never pooled inside a contrast (§3.6), so no test across bodies is available and the disagreement is described but not established. On myoFinger the content-specificity ordering runs opposite to elbow's, and we do not have an account of it. Both cells are post hoc and are ratios of sample variances rather than paired-difference contrasts. Two further mundane rivals are reported rather than argued away: the CV ratios are smaller than the SD ratios, so the effect is partly a consequence of the mean differences reported separately. And a shared-artefact rival applies to both routes equally: all twelve coach seeds are distilled toward one frozen teacher and all twelve prior seeds are handed one frozen world-model checkpoint, so contraction is partly mechanical in both.

H The remaining limitations

These are the eight §6 defers. Each is stated in full; none of them is load-bearing for a claim in §4 or §5, which is why §6 carries the other seven instead.

Claims withdrawn during the study. Five claims were withdrawn over the course of the work, in every case after the control or the direct paired contrast that tested them was run: a double dissociation between the two routes, a back-loading of the prior’s advantage, a withdrawal asymmetry between routes, a decay of the prior’s contribution to the combination, and a dispersion "inversion" that is a null (§4.8). Separately, an interval-estimator error, a normal-approximation multiplier ($z = 1.96$) where a t multiplier is required at $n=12$, was corrected by recomputing all 132 contrasts under the policy of §3.6, a correction that could only remove significance and never create it. The itemised record, every withdrawal, every recomputation, the scope and body-count corrections behind §4.10 and Appendix F, and the unarchived build transcripts of §3.3, is in `RECOMPUTED.md`, and any completeness statement about this repository is scoped there to *no unreported artefact remains that writes a file into this tree*, with named exclusions: thirty-two further 12k-step runs sit under `actors_withdraw/` and `actors_randcoach/` (twelve, at two seeds), `actors_finger/` (sixteen, four arms $\times$ four seeds on myoFinger, logs in `logs_fingervid/`) and `actors_video/` (four, myoElbow seed 0), all trained after this analysis was complete solely to render the comparison videos. Twenty-seven re-execute conditions already reported, bit-identically where we have checked (video-run finger `blank` at 12k is 78.4 / 107.8 / 142.7 / 106.8 against the study’s 78.4247 / 107.8350 / 142.7321 / 106.8157). The other five are the study’s only held-out scoring of `randprior`: one on myoElbow at seed 0 (`actors_video/randprior_actor.pt`, evaluation output not redirected to a file and unrecoverable) and four on myoFinger at four seeds: `randprior - blank` = +37.30 [−31.00, +105.60] and `randprior - prior` = +63.70 [+11.26, +116.14]. They were not planned as an arm, the finger four are $n=4$, and they carry no claim.

- The obvious rival reading of the myoHand null is one these data cannot settle. A body model might be expected to pay off *more* on a harder body, not less, and myoHand is the test: 23 joints and 39 muscles against myoFinger’s 4 and 5. But neither arm improves there within 60k steps, so there is no advantage to amplify and nothing to measure. Whether the larger bodies fail because the method does not scale, or because the budget does, is a question this study cannot separate. The literature it opens against trains for hundreds of millions of steps and the hardest body here received 6×10^4 . The hypothesis is untested at this budget and is not refuted by it.
- The frozen world model is a single checkpoint per body, loaded frozen and handed identically to all twelve seeds, so every `prior`, `randprior` and `priorcoach` interval is conditional on one artefact whose seed variability is unquantified.
- Seven analysed conditions (fourteen directories) have no archived launcher (`results_noanneal_*`, `results_abruptcoach_*`, `results_cleancoach_*`, `results_cleancoach_train_*`, `results_pcconst_*`, `results_prioroff_*`, `results_priorpurge_*`). Their invocations are reconstructible from the run log and checkpoint structure but not from a committed script. Two are provable from the data itself on myoElbow, where the constant and abrupt coach arms agree to ≤ 0.02 mrad before the cut and a teacher swap costs ≈ 5.8 mrad at 2k (on myoFinger the same pair differs by up to 1.67 mm pre-cut, small beside the +46.37 mm withdrawal cost but not a bit-identity), but the constant coach’s teacher identity relative to the annealed coach rests on the run log alone, and §3.2’s `coach(const) - coach(anneal)` contrast depends on it.
- There is one teacher per body (seed 100), trained with 30,000 rewarded task-specific steps that no other arm receives and saved as a terminal checkpoint off a non-monotone curve with no model selection, so the teacher-ceiling account of §4.5 is untested and every teacher-dependent claim is scoped to the teacher as captured.
- All twelve seeds share one training target set and one held-out target set, so held-out generalization rests on a single draw of 20 targets and the dispersion analysed in §4.8 contains no target-set variance at all.

- No hyperparameter search was run, so baseline-tuning parity between **blank** and the aided arms is assumed but not established, and the omission is not neutral between the routes, since the coach has one free parameter against the prior’s five untuned Dyna knobs.
- No behavioural variable beyond distance and return was logged, no activation magnitude, path length, terminal velocity or per-target error distribution, which is the cheapest missing diagnostic in the study.
- The guidance route’s adversarial control was added late (§3.2, §4.2): **randcoach** runs from step 0, so it excludes "any fixed, non-flailing action target lowers early error" for the kept aid. It was not run as a *swap at 8k*, so the withdrawal cells of §4.4 still have no matched random-teacher comparison.

I Protocol and implementation details

The material §3.1, §3.3, §3.4 and §3.5 defer, verbatim.

Evaluation protocol. Actions are greedy (deterministic); the start pose is fixed; each episode is 100 environment steps, one episode per target; there are 8 training targets and 20 disjoint held-out targets; the reported distance is the final-step distance averaged over targets, and teachers, students and referents are scored identically. `evaluate()` in `run_reach.py` logs exactly two quantities per checkpoint, and both are used in this paper. `eval_dist` is that reported error, the final-step target distance averaged over targets, in mrad on `myoElbow` and mm on `myoFinger`. `eval_return` is the undiscounted sum of the environment’s per-step task reward over the 100 steps of the same greedy episode, averaged over the same targets. On `myoFinger` the per-step reward is $r = -\text{dist} + 0.1 \cdot \mathbb{1}\{\text{dist} < 10 \text{ mm}\} - 0.001 \cdot \Sigma \|\text{ctrl}\|^2$ (`myofinger_env.py`), so `eval_return` is in the same length unit as `eval_dist` multiplied by steps, and $-\text{eval_return}/100$ is a mean within-episode error in `eval_dist`’s own units, up to the small target bonus and control-effort terms. §5’s movement index is exactly the difference between that mean and the terminal error. The target sets are fixed global constants shared by all twelve seeds. The run seed enters only through network and environment initialization and the order in which training targets are sampled at reset, so held-out generalization rests on a single draw of 20 targets and the seed-to-seed dispersion of §4.8 contains no target-set variance.

Backbone. One SAC backbone, hidden width 128, update-to-data ratio 2, LayerNorm critics, gradient clipping (norm 10), entropy target $-0.5 \cdot |A|$, lr $3e-4$. Dyna (used by `prior`, `randprior`, `priorcoach`): rollout horizon $k = 3$, `dyna_every` = 100, `dyna_starts` = 200, batch 128, `real_ratio` = 0.5. This is the single set of defaults in `run_reach.py`, fixed once at the start of the study and applied unchanged to every condition, seed and body that carries analysis. The `myoArm` probe of Appendix F used `hidden=256` and `MyoSuite`’s native reward instead, and the `colearn` arm of §4.7 deviates in the ways enumerated there, a different learning rate, re-initialisation at every refresh, refitted normalizers, a validation holdout and a periodic RNG reset. No hyperparameter search was run at any point.

The prior is an MLP $(s, a) \rightarrow \Delta s$, trained on task-agnostic random (Ornstein–Uhlenbeck) muscle activations with no task and no reward, then reused frozen. Five-step open-loop accuracy: `myoElbow` 19.34 mrad (1.108°) joint error; `myoFinger` 14.3 mm mean, 9.8 mm median 5-step fingertip error; `myoHand` 8.5 mm mean, 7.5 mm median (on 3×10^5 babble transitions). The archived build transcripts give 2×10^5 transitions for `myoFinger` (39 s of collection on CPU) and 3×10^5 for `myoHand` (31 min), and the elbow budget is 10^5 , so the three bodies run 10^5 , 2×10^5 and 3×10^5 . The elbow prior still has no archived build transcript, and `prior_myو.pt` is on disk and is loaded by every elbow `prior`, `randprior` and `priorcoach` run in the study with no log of its babbling or of its open-loop evaluation anywhere in the repository. The budget is nonetheless recoverable from the artefact. `arm_env.py` dispatches on `WM_BODY`, so `build_prior_sizes.py`, the committed script behind the archived toy-arm sweep, runs on `myoElbow` unchanged, and its 10^5 model is bit-identical to `prior_myو.pt`: equal element by element across all 71,688 parameters, with the four stored normalizer statistics matching. The neighbouring budgets are not close, 3×10^4 giving 0.02518 rad at five steps and 2×10^5 giving 0.00766 against the checkpoint’s 0.01934, so the identification does not rest on a narrow margin. Three limits travel with it. A reproduction is not a transcript: it records what a committed script at its own defaults produces today, not what was run on the day. The sweep fixes seed 0, hidden 256 and 40 epochs because the script does, so it shows those settings suffice, not that they were chosen. And

it says nothing about the fidelity figures or their scope. The fidelity figures are themselves recoverable the same way: re-measured under this protocol, `prior_myo.pt` gives 19.34 mrad (1.108°) at five steps on the babble distribution, and on the on-policy state distribution 24.86 mrad (1.424°) at five steps, 10.37 mrad at three steps against 10.94 mrad on babble, and 29.90 mrad at six; the persistence baseline at five steps is 292.26 mrad on babble and 245.68 mrad on-policy. That column’s rollouts are driven by a single-seed elbow prior agent, so it is an $n = 1$ policy, and only myoElbow was re-measured, so the myoFinger and myoHand figures remain babble-distribution measurements and remain an optimistic bound on fidelity in the regime where the model is queried. Both reproductions ship: `build_prior.py` and `build_prior_sizes.py`, which also produced the archived myoFinger and myoHand figures, and the two-distribution measurement script, are committed alongside the analysis scripts, with the output of each run in the same tree. One checkpoint per body is loaded frozen and handed identically to all twelve seeds.

Teachers are trained from `blank` (model-free SAC, no body prior anywhere) for 30,000 rewarded, task-specific environment steps, $2.5\times$ a student’s entire budget, at a single seed (100) per body. `train_teacher2.py` writes exactly one checkpoint, at the end of training, with no model selection. Held-out teacher scores: elbow 27.2 mrad, far better than any student; finger 169.1 mm, worse than every student condition mean bar one, the withdrawn finger coach (169.14) being numerically identical to it and just under the no-op referent of 172.3 mm. Both saves come off violently non-monotone curves: the elbow save lands on the global minimum of its own train-target curve (which spans 15.5–149.0), while the finger save (133.5 on train targets) is worse than each of the three checkpoints before it, on a curve spanning 102.9–207.9. The 12k checkpoints of the same two runs, scored with the unmodified `teacher_eval.py`, reach 62.6 mrad and 129.4 mm held-out. Every claim resting on teacher quality is therefore scoped to *the teacher as captured*: the seed-100 checkpoint at exactly 30,000 steps. Teachers produced by invoking the `prior` condition carry the body prior in laundered form. All coach and prior+coach cells reported here use clean teachers, with one labelled exception: Appendix K quotes the contaminated 2×2 coach’s train-target endpoint (+9.10) beside the clean annealed cell, because the constant schedule was never run on train targets at all. The decontamination contrasts are in Appendix D.

Neither intervention arm is matched to `blank`, and the two routes are not matched to each other. The prior arm consumes task-agnostic, reward-free babble transitions of the least task-relevant kind before its 12k-step run, and the archived counts are the ones quoted in §3.3: 2×10^5 transitions on myoFinger and 3×10^5 on myoHand, with no archived transcript at all for myoElbow. The prior arm’s babble budget is therefore $6.7\times$ the 30,000 steps the coach’s teacher consumes on myoFinger ($2\times 10^5 \div 3\times 10^4 = 6.67$) and $3.3\times$ on myoElbow ($10^5 \div 3\times 10^4 = 3.33$), the finger figure from two archived transcripts and the elbow one from an archived transcript and a reconstructed budget. The coach arm’s teacher consumes 30,000 rewarded steps *on this very reach task*, which is directly about the scored quantity. The prior arm is thus simultaneously ahead on raw interaction, behind on task-relevant interaction, and behind on real-data gradient signal by $2\times$ per update (§3.1). No experiment here equalizes any of the three.

I.1 Withdrawal mechanics and the statistical rule

The two paragraphs §3.4 and §3.6 defer, verbatim.

Each aid is cut at env-step 8000 and every condition is scored at 12000. For the coach this requires the abrupt schedule. For the prior we distinguish soft withdrawal (stop generating imagined transitions but continue sampling those already buffered) from strict withdrawal (stop generating *and* purge the buffer). Strict was intended as the matched control, for a design reason: the coach condition has no imagined-transition buffer at all, so for the prior arm to be matched, removal must also leave the agent training on real data only. As implemented it is not matched, and the mismatch is in exactly that respect. Purging sets the imagined buffer’s count to zero, and rollouts also stop at the cut, so the buffer never refills. The update rule’s guard (`model_buf.n < batch`) is then permanently satisfied and the strict arm silently switches from 64 real + 64 imagined to 128 real per update for the rest of training. The prior’s strict withdrawal therefore changes two things at 8k, imagination is removed *and* the real-data gradient signal doubles, while the coach’s abrupt withdrawal changes one. The soft arm keeps the mix at 64 + 64 but on a frozen, stale imagined buffer, so it is unmatched in the other direction. Neither withdrawal is a clean single-factor manipulation, and we report both with that stated.

The primary statistic is the paired t 95% CI, $\text{mean}(\mathbf{d}) \pm t_{0.975, n-1} \cdot \text{sd}(\mathbf{d})/\sqrt{n}$, reported as an effect size with its interval. A contrast is reported as significant only when that interval excludes zero. Two robustness checks accompany it and neither is used to license a claim: a BCa bootstrap 95% CI (20 000 resamples, RNG seed 0, resampling the vector of paired seed differences) and an exact Wilcoxon signed-rank test. Two re-scalings of the decay interaction are declared here, used only in §4.5 and §7 and licensing no separate claim. `[coach(const) - prior]@LATE - @EARLY` is a difference of differences of a quantity bounded below by zero, measured at two very different error levels, so it is reported under three scalings: absolute, $(\mathbf{c_L} - \mathbf{p_L}) - (\mathbf{c_E} - \mathbf{p_E})$, in mrad or mm; log-error, $(\ln \mathbf{c_L} - \ln \mathbf{p_L}) - (\ln \mathbf{c_E} - \ln \mathbf{p_E})$, a change in the *ratio* between the two arms; and blank-normalised, each arm divided by that seed’s own **blank** in the same window before differencing, so the units are fractions of the untreated baseline. All three use the paired t interval above. Where they disagree, §4.5 reports the disagreement and does not choose among them. Over the 134 contrasts computed, 60 are significant under t , 60 under neither estimator, and 14 under BCa alone. The t -only cell is empty, so BCa is uniformly the more permissive side. The 14 BCa-only contrasts are tabulated in Appendix A and carry no claim. Per-checkpoint cells sit outside every correction family reported here, because they sub-divide windows already in the family and correcting them would need a scheme we have not run. The per-checkpoint finger cells of §4.5 are therefore asserted uncorrected, and §4.10 records that the exemption was kept even where entering the cells would have helped us. All contrasts are computed on unrounded per-seed data and may differ from differences of displayed rounded means by ~ 0.1 unit. The full recomputation of every contrast outside the **randcoach** arm is **RECOMPUTED.md**. Its 134-contrast register carries both estimators. The contrasts added later carry the primary estimator only: the two non-absolute scalings of the decay interaction per body (§4.5), the frozen12 cells (§4.7), every myoHand cell (§4.10), the train-target coach cells (§4.3) and the per-checkpoint cells (§4.5), as do the **randcoach** contrasts of §4.2, which `analyze_randcoach.py` computes.

J The remaining next steps

§7 names the first four and, out of order, **blank** to 30k. This is the full list as it stood at submission.

Next steps. Log at least one behavioural variable, mean activation, path length, terminal velocity, per-target error, at every evaluation checkpoint, and resample the target sets per seed. Then, in rough order of cost, we would run the swap half of the **randcoach** control, whose from-step-0 arm is reported in §4.2. Substituting a random teacher *at the 8k withdrawal point* would separate the teacher’s content from any fixed non-flailing target in the withdrawal cells, and would separate the withdrawal cost from the optimization shock. After that we would add a matched-replay Dyna control that swaps the imagined half of each batch for replayed real transitions, and run **warm** and **colearn** at $n=12$ on myoFinger and on held-out targets; the myoElbow train-target half is now run (§4.7), and the remaining halves would test whether the tight null and the babble/dispersion effect replicate. A withdrawal experiment sized against the *direct cost-difference* contrast would need more seeds than we ran. Multi-seed and multi-checkpoint clean teachers, since running a student against the 12k teacher checkpoints already on disk is a teacher-quality manipulation available at zero training cost. Intermediate distillation weights. **blank** to 30k at $n=12$, which would settle the amortization reading. A matched-total-interaction baseline. And controls breaking the information asymmetry in both its parts: a task-agnostic coach or a task-aware body model, and a Dyna variant with a *learned* reward model in place of the analytic oracle.

K The forward-model qualifications, the amortization series and the movement index

The full versions of the material §4.2, §4.3, §4.4, §5 and §7 summarise.

The two routes can now be compared directly. While present, the coach is the stronger aid at the myoElbow held-out endpoint and faster than the prior on both bodies. That lead decays on both, and inside a 12k budget the decay crosses zero on myoFinger and not on myoElbow. The route comparison is the contrast most exposed to the design asymmetries of §3.1 and §3.5. The coach arm trains on twice the real data per update, while the prior arm consumes far more raw interaction ($6.7\times$ the teacher’s 30,000 steps on myoFinger

and $3.3\times$ on myoElbow, §3.5) and holds a reward oracle the coach does not. When both aids are present we cannot detect a difference between the combination’s endpoint and the coach alone’s on either body, and after a matched withdrawal no route-level *durability* statement is available. Every claim this paper makes about an aid beating model-free RL at the endpoint is an elbow claim: on finger the prior-versus-blank cells are not significant under the primary estimator, the constant coach beats neither **blank** nor the prior at 12k, and the single finger endpoint claim against model-free RL runs the other way (the withdrawn coach is +51.5 worse). That restriction to the endpoint is load-bearing, because the EARLY window runs the other way: on finger both coach schedules beat model-free RL early (−17.60, −17.39).

Four qualifications belong in the same breath. The prior arm evaluates the environment’s analytic reward at its imagined states, so the endpoint reversal is scoped to a forward model *plus a known reward function* and is as much an oracle result as a forward-model result (§3.5). Cutting the prior is not shown to be costly on either body and the direct removal-cost difference between the routes is non-significant on both, a null with a named candidate generator (§3.4), so nothing here says that what the model contributes is better retained. The decay holds in raw units and under blank-normalisation on both bodies and, on myoElbow alone, not under the log-error reading (§4.5). And the reversal’s status under multiplicity correction is family-dependent: it survives four of the six definitions we report, including the one §3.6’s own exclusion criterion derives, and fails the unpruned literal set, while seven cells survive every definition in which they appear (§4.11).

coach(const) – prior in the EARLY window {2,4,6k}, held-out, is −50.56 on elbow and −11.72 on finger: the constant-weight coach is ahead of the frozen prior early on both analysed bodies. That lead is bought by what the teacher contains. The adversarial control for this route, **randcoach**, a random-initialized teacher of identical architecture in the same loss at the same weight, does not reproduce that gain on myoFinger and is significantly worse than no teacher on myoElbow. The direct within-axis contrast **coach(const) – randcoach** is −19.03 [−24.40, −13.67] on finger and −108.60 [−142.01, −75.18] on elbow; against model-free RL the same arm is +1.43 [−6.03, +8.89] on finger EARLY and +24.41 [−3.16, +51.99] at the endpoint, both failures to detect. The finger endpoint twin is −19.25 [−32.66, −5.84]. No **randcoach** contrast appears in RECOMPUTED.md; they are computed by **analyze_randcoach.py** under the primary estimator only (§3.6). §5 records why this control is the one the guidance claim needed. Against model-free RL in the same window, where the aid is unambiguously present: **coach(const) – blank** elbow −59.03 [−87.11, −30.95], finger −17.60 [−23.93, −11.27]. The tapered coach reproduces the same EARLY effect on both bodies (elbow −48.27, finger −17.39). The prior’s own EARLY cell against model-free RL is not significant under the primary estimator on either body (elbow −8.47 [−31.82, +14.88]; finger −5.88 [−16.15, +4.38], BCa-only, Appendix A, the same class §4.3 flags for its siblings), and it decomposes the elbow route contrast exactly: −59.03 + 8.47 = −50.56. On elbow in the EARLY window, therefore, **coach – prior** is **coach – blank** plus a quantity we cannot distinguish from zero. This was not the a-priori favourite: the frozen model supplies exactly what classically pays off earliest (Janner et al., 2019; Chua et al., 2018), and the two EARLY nulls above are where that expectation fails (§1).

The prior’s elbow effect is $70.7 \rightarrow 50.0$ mrad held-out, -20.7 mrad $\approx -1.19^\circ$, a $\sim 29\%$ reduction in reach error, significant on held-out and on train targets. The coach’s endpoint advantage has no train-target twin, because the constant schedule was never run on train targets. The two schedules that were run there, annealed and the contaminated 2×2 coach, do not beat the prior on train targets (+15.24 and +9.10 [−12.61, +30.80], both n.s.). Their point estimates run opposite to the held-out −10.89. Both train-target coach arms use the default linear taper, so their coach weight is already zero from 8k to the 12k scoring point. They are post-withdrawal numbers and not a still-attached twin. We report that asymmetry rather than leaving the prior’s corroboration row to stand alone. The constant-weight coach improves elbow endpoint accuracy by more (−31.6 against model-free SAC), and the direct route-versus-route contrast, the quantity a claim about which aid is stronger actually requires, is −10.89 in the coach’s favour. While present, the coach is the stronger aid at the myoElbow held-out endpoint. Nothing is claimed for the prior on finger: the three prior-versus-blank finger cells of Appendix A are significant under BCa but not under the primary estimator, and the finger MID and LATE cells (Appendix D, −1.32 and −12.56 [−37.94, +12.81]) are significant under neither. Two scope conditions travel with these cells. The constant coach’s endpoint is scored while the loss is still pulling the student toward a competent teacher, so it is close to definitional taken alone and

what makes the pair informative is the contrast *between schedules*; and the two arms are not scored on equal real-data gradient signal (§3.1).

Cutting the coach is costly on both bodies. On myoElbow, where the teacher as captured is competent, that cost is a within-route acquisition-versus-retention dissociation and is the cleanest result in the withdrawal experiment; §5 reads the finger cost differently, since the finger teacher sits essentially at the no-op referent. On finger, removing the coach leaves the agent worse than never having had one (+51.54 against **blank**) and worse than the withdrawn prior (+64.45), both produced by the withdrawn finger coach landing essentially at the no-op referent rather than by the withdrawn prior being retentive. Cutting the prior is not shown to be costly on either body, and no route-level durability asymmetry is claimed: the prior’s own withdrawal cost is non-significant on both bodies (**prior strict-withdrawn** – **prior kept**: elbow +7.17; finger +14.19), and so is the direct route comparison (**coach cost** – **prior cost**: elbow +8.83; finger +32.18). Neither null is treated as evidence of retention, and the first has a named candidate generator: by §3.4 the strict arm doubles its real-data gradient signal at the cut, which should *help* it, biasing this very contrast toward zero, and the finger interval misses zero by 0.95 mm. The derived **coach cost** – **prior cost** contrasts inherit the same bias, in the direction of inflating the coach’s cost relative to the prior’s. The elbow soft-withdrawn prior does beat **blank** (–18.62), but that manipulation is unmatched and biased in the prior’s favour, so it is not evidence that the prior persists. The cost does not identify two things: cutting the coach at 8k removes the teacher’s information *and* deletes a weight- ≈ 1.0 auxiliary loss term mid-training, and Appendix O reads the finger cost as dominated by that optimization shock. Neither prior withdrawal is single-factor either (§6).

That leaves the controls, and what the guidance control settled. The two routes are not equally well controlled. The forward-model route has an adversarial control, **randprior**. The guidance route had none for most of this study. **randcoach**, a randomly-initialized frozen teacher distilled with the same auxiliary loss and weight, was added last. That gap matters most exactly where the guidance result looks strongest. The clean finger teacher as captured scores 169.1 mm, worse than a blank student and within 3.2 mm of the no-op referent, and yet the finger coach still produces a large EARLY lead. So "distil the student toward a near-no-op action target on the body where inaction is worth 71 mm over thrashing" predicts an early gain of the observed sign *independently of anything the teacher encodes about the task*. Without a control, that account could not be separated from ours. We have now run the control. **randcoach** distils a randomly-initialized frozen teacher of identical architecture through the same auxiliary loss at the same constant weight, matched to **coach(const)** in body, budget, target sets and seeds (§3.2). On the body that raised the rival account, the direct contrast separates the two. On myoFinger EARLY, **randcoach** – **coach(const)** is +19.03 [+13.67, +24.40]. That is the within-axis comparison §1 requires, and does not infer one from two separate verdicts. A randomly-initialized teacher is itself a fixed non-flailing action target, and it does not reproduce the coach’s early gain. Against model-free RL the same arm is +1.43 mm [–6.03, +8.89], where **coach(const)** – **blank** is –17.60 [–23.93, –11.27]. Per §6 that interval is a failure to detect at $n=12$ and is not evidence that a random teacher does nothing. On myoElbow an uninformative teacher is significantly *worse* than no teacher at every window (EARLY +49.57 [+23.37, +75.76]; MID +21.54 [+1.53, +41.54]; LATE +20.29 [+0.58, +40.00]; ENDPOINT +22.53 [+2.80, +42.26]). That is the shape **randprior** shows for the forward-model route on the same body at the endpoint, the only window on which it was scored. Both routes now carry an adversarial control, and on myoElbow each is worse than no aid on the target set it was run on: **randprior** on train (+23.8 [+1.7, +45.8]), **randcoach** held-out (+49.6 [+23.4, +75.8]). The two controls are parallel in design and are not matched in scoring. On myoFinger **randcoach** is not distinguishable from model-free RL at EARLY or at the endpoint, and **randprior** is not distinguishable from it at the endpoint on train targets, the only window and target set on which it was scored; **randcoach** is significantly worse than model-free RL at MID (+12.86 [+8.12, +17.60]) and LATE (+26.44 [+11.86, +41.02]), the same shape it shows on myoElbow at every window (§4.2, §4.6).

Four qualifications follow from our own controls. First, the *shape* is unusual: on elbow the benefit is not detected EARLY (–8.47 [–31.82, +14.88], an interval that does not exclude an effect the size of the endpoint one), not detected in MID (–12.25 [–25.34, +0.84]; BCa-only, Appendix A, and so carrying no claim), and present LATE and at the endpoint, the reverse of the classical model-based story (Janner et al., 2019; Chua et al., 2018). That shape is three window verdicts, not a tested growth. The within-seed interaction §3.6

referent	elbow train (mrad)	elbow held-out (mrad)	finger train (mm)	finger held-out (mm)
random-action policy	634.9	659.8	220.0	243.4
zero-activation (no-op)	676.2	692.4	155.2	172.3

Table 13: The competence referents, scored under the identical protocol and target sets.

requires for any claim that an advantage grows, `[prior - blank]@LATE - @EARLY`, is null on both bodies, elbow -13.45 $[-39.34, +12.45]$, finger -6.68 $[-36.49, +23.13]$, so the pattern is described here without being established. The corresponding `coach - prior` interactions, which are significant, are the ones §4.5 rests on. One plausible but untested mechanism is that the frozen model becomes useful once the policy’s state distribution has expanded. A second, entirely mundane one is that this arm is handicapped by 64 real transitions per update throughout, and a handicap of that kind bites hardest early. Second, the arm operationalizes a forward model plus a known analytic reward function evaluable at imagined states, so any biological reading must assume the nervous system has access to something like a state-evaluable cost at unvisited states. That scope covers the finger reversal as much as the elbow accuracy result: an arm that can evaluate the true reward at unvisited states has a standing mechanism to keep improving after model-free arms plateau, and the finger prior falls $107.31 \rightarrow 90.50$ over exactly the last 2k that carries most of the $+32.27$. Read that way the reversal is an oracle result as much as a forward-model result, and nothing here separates the two. Third, the elbow model’s own 19.3 mrad open-loop prediction error is essentially the size of the -20.7 mrad endpoint effect attributed to its fidelity. That figure has, as §3.3 records, no archived build transcript behind it, but the checkpoint survives and reproduces it on re-measurement, and it is no longer a babble-distribution figure alone: on the on-policy distribution where the model is actually queried the error is 24.9 mrad at five steps and 10.4 mrad at the three-step horizon the Dyna loop uses (Appendix I).

Fourth, both endpoint accuracy results have an amortization reading that the available data do not exclude. Four unaided long-budget elbow runs sit on disk. At each run’s true terminal checkpoint, 32k for the three continued `blank` seeds and 30k for the teacher run, they end at 31.5, 35.3, 41.8 and 27.2. Three of the four are below the constant-coach student’s 39.13, and all four are below the kept prior’s 50.02, by a wider margin in the prior’s case. The same probe on `myoFinger` points the same way against the coach and not against the prior: at 32k the three continued seeds are $118.4 / 87.2 / 104.8$, plus the teacher run’s 169.1 at 30k, three of the four below the constant coach’s 122.77, while only one of the four falls below the kept prior’s 90.50. Under that reading, on `myoElbow`, each aid buys a place on the `blank` learning curve, and does not reach something a longer unaided run would miss. Each of the four is a single terminal checkpoint of an oscillating curve with no interval, and the settling experiment, `blank` to 30k at $n=12$, was not run. One thing does constrain it. The archived `eval_return` makes a movement index recoverable on finger, $M = (-\text{eval_return}/100) - \text{eval_dist}$, whose sign separates "ends nearer than its episode average" from "ends farther". Scored on §4.9’s own referents under the identical protocol, the zero-activation no-op is at $M = -40.12$ held-out (-26.89 train), random action -47.03 and the clean finger teacher -17.69 . All eleven finger arms lie in $[-8.16, +2.80]$, each at least 29.7 mm from the no-op referent measured on its own target set. Per-seed SDs run 2.55–11.28, so the separation is $\approx 7.5\sigma$ at the arm with the smallest separation in millimetres and 2.8σ at the noisiest, `randcoach` ($M = -8.16$, per-seed SD 11.28, 31.96 mm from the held-out no-op referent), which was added late and is computed here rather than in `RECOMPUTED.md` D1. No finger arm reproduces the zero-activation policy’s movement signature. What M cannot do is measure how *much* an arm moves, or discriminate against a constant-activation policy other than the zero-activation one, which is what the rival actually turns on. The `randcoach` control above settles the rival directly. A single logged mean activation would still sharpen the mechanism.

L Competence referents, the long-budget probe, and `myoHand`

The two subsections §4.9 and §4.10 summarise, in full.

Table 13 gives the absolute competence referents, scored under the identical protocol and target sets.

step (k)	12	14	16	18	20	22	24	26	28	30	32
elbow blank − prior (mrad)	+4.1	+17.8	+2.7	−9.9	+15.1	+30.9	+10.9	−7.1	−5.2	−4.1	−5.4
finger blank − prior (mm)	+26.5	+53.2	+38.8	+52.8	+35.0	+44.6	+30.4	+40.1	+38.6	+67.5	+43.8

Table 14: Both **blank** − **prior** long-budget series in full, at every archived checkpoint.

These are unreplicated point estimates: `random_baseline.py` runs one rollout per target under one RNG seed. Re-run over eight RNG seeds on finger, the random-action referent is 249.2 ± 10.8 mm held-out and 247.4 ± 14.2 mm train, so the archived 220.0 mm train figure is a favourable draw. The elbow referent was not re-drawn. We use solved in exactly one sense: a body is solved by a condition when that condition’s held-out error is an order of magnitude below the random-action referent and far below the no-op referent. Only one of the two bodies gives a resolvable ordering between the referents. On elbow random-action (659.8) is nominally below no-op (692.4), a gap of 32.6 mrad against across-target SDs of 460 and 543 mrad on those two draws, roughly 0.2 standard errors, and the elbow referent was not re-drawn, while re-drawing the finger referent under the identical protocol moved it by 12.5% of the archived draw ($220.0 \rightarrow 247.4 \pm 14.2$, itself 1.9 SD above the archived value), so we report the elbow numbers and read no ordering from them. On finger the gap is far larger and runs the other way: no-op 172.3 beats random-action 243.4 by 71 mm, so on myoFinger inaction is a good policy and thrashing an actively bad one, and the clean finger teacher as captured (169.1 mm) sits 3.2 mm below the zero-activation referent. Distilling toward it is therefore close to distilling toward "stop moving" on the body where stopping is good (§5).

Eight of the ten elbow conditions are solved in that sense. Model-free **blank** ($9.33\times$) and the random-teacher control **randcoach** ($659.8/93.26 = 7.07\times$) are not. Ratios against the 659.8 referent: **priorcoach(const)** $17.1\times$, **coach(const)** $16.9\times$, **prior kept** $13.2\times$, **prior soft** $12.7\times$, **coach abrupt-withdrawn** $12.0\times$, **prior strict-withdrawn** $11.5\times$, **priorcoach(anneal)** $11.4\times$, **coach annealed** $10.8\times$, and **blank** at 70.73 mrad is $9.33\times$, which is not an order of magnitude, a miss of 6.7%, well inside the imprecision of a single unreplicated draw. Every elbow contrast in this paper is a difference *within* that regime, on a 1-DoF body whose held-out targets are near-interpolation. myoFinger is solved by no condition: the two lowest-error finger conditions are the soft-withdrawn prior at 88.1 mm and the kept prior at 90.5 mm, only $\approx 1.9\times$ below doing nothing, and model-free SAC (117.6) is only $\sim 32\%$ better than doing nothing. A held-out fingertip error of 90.5 mm is comparable to the finger’s own length scale, so the finger endpoint result says which of two poor conditions is less poor.

A supplementary longer-budget experiment (**blank** and **prior**, $n=3$ seeds, curves parsed from logs, no intervals) supports no claim. The runs were launched for 36k and killed at 32k, except **prior** seed 2, whose log stops at 30k. The 12k–30k columns are paired at $n=3$, and the 32k column alone is unpaired, **blank** over three seeds against **prior** over two. Table 14 prints both **blank** − **prior** series in full.

Both series oscillate. On elbow the gap changes sign three times and **blank** is ahead of the prior at 18k, 26k, 28k, 30k and 32k. On finger the gap is positive at all eleven checkpoints and shows no trend after 14k. We read the two halves symmetrically. The elbow half is used below as reason to doubt the 12k elbow effect. The finger half runs the other way. The prior is ahead of **blank** at every checkpoint out to 32k, the same direction as the 12k **prior** − **blank** finger contrast that $n=12$ could not resolve (-27.10). That consistency is not evidence: $n=3$, no intervals, checkpoints within a run are strongly autocorrelated so eleven of them are not eleven observations, and the finger **blank** series itself swings between 95 and 134 mm over the same stretch. We report it because using this probe only where it costs us, and skipping it where it does not, would be a selection, and the correct conclusion from it on both bodies is the same one: it supports no claim. These three seeds are seeds 0–2 of the main study, bit-identical through 12k, and their own 12k **prior** − **blank** gap is -4.2 mrad against the 12-seed -20.7 (per seed -23.8 , $+10.6$, $+0.7$) at a **blank** mean of 57.8 against 70.73, a favourable subsample of the baseline in which the elbow effect had already disappeared *before* the extra budget was spent. They are direct reason to doubt that the 12k elbow effect survives to 30k, and no long-budget run was done with a coach of any schedule.

Four MyoSuite bodies were instantiated and trained on in the course of this study. Two carry the analysis, this subsection reports the third, and the fourth, myoArm, a single truncated **blank** seed under MyoSuite’s

own reward, is in Appendix F. Neither the third nor the fourth carries a claim. `results_hand/` holds twelve complete runs on myoHand (39 muscles), `{blank, prior}` $\times$ six seeds, thirty checkpoints each, 60,000 environment steps, train targets, produced by `launch_hand.ps1` calling the same `run_reach.py`. We detect no improvement on either arm over the budget. Paired within arm over the six seeds, `blank` at 60k minus `blank` at 2k is +3.88 mm [−7.10, +14.86] ($p = 0.406$) and `prior` is −3.51 [−13.81, +6.80] ($p = 0.422$). On the mean of the last five checkpoints minus the mean of the first five, `blank` is −0.28 [−6.28, +5.71] and `prior` −3.82 [−12.48, +4.83]. The levels behind those intervals: `blank` 102.5 mm at 2k and 106.3 at 60k, `prior` 103.1 and 99.6, both series oscillating flat in the 99–119 mm band for thirty checkpoints. Per §6 an undetected change is a failure to detect, not an absence, so we say that no improvement is detected, and we do not say that neither arm learns.

Paired between arms over the same six seeds, one window cell reaches nominal significance: 12k, `prior` − `blank` = +12.49 [+0.99, +23.99], Wilcoxon 0.03125, 6/6 seeds on the sign. It does not survive the window family §3.6’s criterion would generate for a body with one condition pair and two endpoints. myoHand’s five window cells (EARLY, MID, LATE, the main study’s 12k endpoint and this run’s own 60k endpoint, a set that keeps the MID and LATE cells §3.6’s exclusion criterion would prune, so it is the *unpruned* window family rather than the one that criterion derives) put the 12k cell at Holm rank 1, $p = 0.03832$ against a threshold of $0.05/5 = 0.01000$: it fails, the step-down stops there, and nothing in the family survives. Pruned to $K = 3$ the 12k cell faces 0.0167 and still fails at $p = 0.03832$, so the verdict does not turn on the choice of family. A 30-checkpoint per-checkpoint grid was also computed and returns the same verdict. Two of its 30 cells reach $p < 0.05$ against a chance expectation of 1.5, and Holm over that grid also stops at rank 1. The window family is what §3.6’s policy licenses. The per-checkpoint grid is reported for disclosure only. It is not entered into a correction family, which is the exemption §3.6 applies everywhere else. We keep it even on the one body where entering the grid would have removed an unfavourable cell. The window family is used instead precisely so that the exemption is not granted where it helps and imposed where it hurts. The 12k cell’s neighbours run +8.34 and −10.62, neither significant, and by 60k the sign has changed again (−6.79 [−22.45, +8.86]). The body is uninformative about the forward-model route. It does not refute it, and no claim in this paper rests on it. The window table is in Appendix C.

M Literature detail deferred from §2

The offline model-based distinctions and the guidance-hypothesis evidence, in full.

Offline model-based RL. MOREL (Kidambi et al., 2020), MOPO (Yu et al., 2020) and COMBO (Yu et al., 2021) freeze a pretrained model and then optimize under it, as Plan2Explore’s zero-shot mode also does (Sekar et al., 2020), but they train the model on logged interaction data and constrain the optimization, by uncertainty-based pessimism (Kidambi et al., 2020; Yu et al., 2020) or by conservative value regularization (Yu et al., 2021), COMBO having been built specifically to avoid uncertainty estimation. Only MOREL optimizes purely inside the model. MOPO and COMBO both draw policy-update batches from the logged data as well. We freeze a model learned from reward-free, task-agnostic babbling and use it optimistically. Unlike that family, our prior arm is also handed the environment’s analytic reward evaluated at imagined states (§3.5), task information MOPO and COMBO learn from data, their dynamics models predicting reward alongside next state, though MOREL likewise assumes a known reward function (§3 of (Kidambi et al., 2020)).

Guidance, augmented feedback, and policy distillation. The guidance hypothesis (Salmoni et al., 1984) holds that augmented feedback guides the learner toward the correct response while breeding a dependency on it, so frequent feedback can flatter acquisition relative to retention. We state it that way instead of as the familiar "helps acquisition, hurts retention", because its best-known test supports only the retention half. Winstein & Schmidt (1990) report a reliable benefit of reduced feedback at *delayed* no-feedback retention under a faded schedule, while acquisition was not reliably degraded. In their Experiment 2 the reduced-feedback group was significantly *better* during acquisition, but only on the second practice day and only when scored on feedback trials. On Day 1 no group difference appeared even there, and the pooled all-trials analysis found no reliable group effect on either day. Their Experiment 1 detected no reliable group difference in either phase, on a 10-minute test rather than a delayed one. The associated *reversal* effect is

meta-analytically weak as well: Marschall et al. (Marschall et al., 2007) report fail-safe N of 2 and 3 against an effect-size threshold of 0.2, and 0 at thresholds of 0.5 and 0.8. The hypothesis’s methodological core is the acquisition-versus-retention distinction: a measure taken *while* the aid is present measures acquisition, and only a test *after removal* measures learning. Our coach is a policy-distillation / kickstarting auxiliary, impersonal guidance and never social scaffolding. We separate the two limbs by running the coach at constant weight (the schedule is defined in §3.2 and the constant-versus-annealed contrast is in Appendix D) *and* cutting both aids at a common step (§3.4, results in §4.4).

N The combination and the frozen-ness experiment

What §4.7 summarises, in full.

`priorcoach(const)` – `prior` reproduces the §4.5 crossover with a different condition pair, EARLY elbow -59.30 $[-80.25, -38.34]$ and finger -16.54 $[-20.62, -12.46]$, ENDPOINT elbow -11.53 $[-17.80, -5.27]$ and finger $+25.63$ $[+8.88, +42.38]$, though non-independently, since it reuses the same runs. Against the coach alone, `priorcoach(const)` – `coach(const)` is non-significant at ENDPOINT on both bodies (elbow -0.64 $[-4.05, +2.77]$, finger -6.64 $[-14.71, +1.43]$): at the endpoint we cannot detect a difference between the combination and the coach alone, which is a null at $n=12$, not a demonstration that combining lands on the coach’s outcome. The elbow EARLY cell is significant (-8.74 $[-17.09, -0.38]$), so the prior is not inert in the presence of a maintained coach on elbow. The matching finger cell is not (-4.82 $[-10.18, +0.54]$). Full combination tables, including the mixture-midpoint contrasts, are in Appendix D.

Frozen-ness is a design commitment carried by the title, and a dedicated $n=12$ experiment tests it directly. `results_frozen12/` holds four arms on myoElbow at the identical 12k budget, twelve matched seeds each on train targets. Two of the four are not new data: `run_reach.py` is deterministic, so the `blank` and `prior` cells re-executed here are byte-identical to their `results_2x2_elbow` originals, and the `prior` – `blank` value of -25.50 mrad is the same number as §4.3’s and is not an independent reproduction. What is new is `warm` and `colearn` at $n=12$. The arms are: `prior` (this study’s frozen babble-learned model), `warm` (the identical babble checkpoint with `freeze=False`, fine-tuned online), `colearn` (a world model trained online from scratch with no babble), and `blank`. `prior` – `warm` isolates freezing cleanly: the two arms load the same checkpoint and differ only in the `freeze` flag. `warm` – `colearn` does not isolate babbling. As implemented, `colearn` also uses a different learning rate ($1e-3$ against $3e-4$), re-initializes its model from random weights at every 2k refresh instead of fine-tuning cumulatively, refits its normalizers on real data, holds out 10% for validation, and, because `train_world_model` reseeds the *global* torch stream that SAC’s exploration draws from, receives a periodic RNG reset that `warm` does not. The re-initialization and the reseeding are each on their own sufficient to generate large seed-to-seed spread, so no causal attribution to babbling is available from this contrast.

Means are `prior` 40.15 mrad, `warm` 39.46, `colearn` 45.66. Freezing is a tight null: `prior` – `warm` = $+0.68$ $[-2.64, +4.01]$, an interval that at 95% confidence rules out effects beyond about $-2.6/+4.0$ mrad, set against the prior’s own -25.50 mrad effect over `blank` on the same target set (§4.3), so freezing neither helps nor hurts at a scale that could matter here. This supersedes the $n=5$ slice in `results_myo/`, where the same contrast was $+1.70$ $[-9.65, +13.05]$, an interval that already excluded a headline-sized (≈ 25 mrad) effect but not a quarter-sized one, which is what the $n=12$ replication narrows. That slice also holds an `imag` arm (a policy trained inside the frozen prior taking zero real environment steps, 44.55 mrad at $n=5$), which the $n=12$ replication does not include. `prior` – `colearn` = -5.51 $[-19.91, +8.89]$ and `warm` – `colearn` = -6.19 $[-19.35, +6.96]$ both remain unresolved, and the second of those is the contrast this design nominates to isolate babbling, an interval about four times wider than the freezing null beside it, so nothing here shows that babbling improves endpoint error.

What separates the arms is dispersion, not accuracy. What produces that dispersion is not identified. Post hoc, on the apparatus Appendix G declares for §4.8: endpoint SDs are `prior` 6.27, `warm` 5.63, `colearn` 19.77. The online-from-scratch model is between about 1.7 and six times as dispersed as either babble-pretrained arm, $SD(colearn)/SD(warm) = 3.51$ $[1.88, 5.83]$, Pitman–Morgan $p = 0.00046$; against `prior`, 3.15 $[1.69, 5.32]$, $p = 0.00077$, while the two babble arms are not distinguishable from each other, $SD(prior)/SD(warm) = 1.11$ $[0.68, 1.76]$. But because `colearn` differs from `warm` in five implementation respects, not only in

step	elbow blank	elbow prior	elbow coach	elbow prior	coach	–	finger blank	finger prior	finger coach	finger coach	–	prior
2k	172.95	136.24	86.60	−49.64 +3.89]	<i>n.s.</i>	[−103.16,	143.26	136.20	134.82	−1.38	[−15.76, +13.00]	<i>n.s.</i>
4k	124.84	120.45	58.43	−62.02 −36.61]		[−87.43,	135.61	133.14	115.39	−17.75	[−25.64, −9.86]	
6k	74.58	90.27	50.25	−40.02 −20.63]		[−59.41,	136.90	128.80	112.77	−16.03	[−22.87, −9.18]	
8k	89.43	60.11	45.52	−14.59 −0.68]		[−28.50,	127.95	130.14	111.26	−18.88	[−33.72, −4.03]	
10k	81.24	58.12	45.35	−12.77 −1.76]		[−23.78,	105.34	107.31	114.95	+7.64	[−3.37, +18.64]	<i>n.s.</i>
12k	70.73	50.02	39.13	−10.89 −2.52]		[−19.26,	117.60	90.50	122.77	+32.27	[+16.49, +48.05]	

Table 15: The myoFinger reversal at all six archived checkpoints, held-out and paired, n=12.

babble-pretraining (above), this study cannot say that the babbling is what buys the reproducibility. The observation is post hoc, on one body and one target set, and carries no numbered claim. Isolating the cause would need `colearn` re-run with matched learning rate, refresh policy, normalizers and RNG handling.

We therefore describe the frozen prior as *the arm we ran*: freezing is a defensible implementation choice that this experiment shows to be inconsequential at this scale, and on the accuracy axis neither freezing nor babbling is shown to matter at n=12. The one place the babble-pretrained arms separate from the online-from-scratch one is seed-to-seed dispersion, which is post hoc and carries no numbered claim.

O The reversal checkpoint by checkpoint

The block §4.5 defers, verbatim.

The reversal, checkpoint by checkpoint. Table 15 gives all six archived checkpoints, held-out and paired, n=12.

Elbow columns are mrad and finger columns are mm. Bold in a mean column marks the two checkpoints at which the numerically lowest of the three is not the constant coach. The finger reversal is directionally consistent but not monotone and is carried by the last 2k: the coach leads at 4k, 6k and 8k (−17.75, −16.03, −18.88), the 10k contrast is +7.64 with a CI containing zero, and most of the +32.27 accrues between 10k and 12k, driven by the prior falling 107.3 → 90.5 while the coach rises 115.0 → 122.8. At 10k model-free **blank** (105.3 mm) is significantly below the constant coach (paired −9.61 mm [−15.62, −3.60]) but is *not* distinguishable from the prior (−1.97 [−15.23, +11.28]; **blank** is the numerically lowest of the three in 6 of 12 seeds), and **blank** itself rises +12.26 mm across 10k→12k, so the stretch carrying the reversal is one in which none of the three curves is behaving like a converged system. We do not localize the crossing more tightly than 8k–12k.

The leading explanation for where the crossing falls is the teacher ceiling of the teachers as captured (§3.4): the clean elbow teacher reaches 27.2 mrad, far better than any student, while the clean finger teacher reaches 169.1 mm, worse than every student condition mean bar one and essentially at the no-op referent, so on finger the student is pulled toward a target worse than what it can find alone. The account is narrower than it looks, because the withdrawal numbers run against the obvious reading of it. If the pull toward 169.1 mm were simply costly, cutting it at 8k should relieve the cost. Instead the kept coach ends at 122.77 and the arm whose coach was cut ends at 169.14, essentially *on* the teacher’s own score. We therefore read the finger withdrawal cost (+46.4) as dominated by the optimization shock §4.4 already names rather than by loss of teacher content, and confine the ceiling account to what it does explain: why the kept finger coach stops improving and rises over the last 4k (111.3 → 122.8). It does not explain the crossover’s main driver either: the decomposition above puts most of the +32.27 in the prior *falling* 107.3 → 90.5, which a ceiling on the coach says nothing about, and that terminal drop is unexplained. The account also rests on two single-seed terminal checkpoints, and no experiment here varies teacher quality while holding everything else fixed.

References

- Berg, Caggiano, and Kumar. SAR: Generalization of Physiological Agility and Dexterity via Synergistic Action Representation. RSS 2023. arXiv:2307.03716, 2023.
- Bongard, Zykov, and Lipson. Resilient Machines Through Continuous Self-Modeling. Science 314(5802):1118-1121, 2006, 2006.
- Caggiano, Wang, Durandau, Sartori, and Kumar. MyoSuite: A Contact-Rich Simulation Suite for Musculoskeletal Motor Control. L4DC 2022. arXiv:2205.13600, 2022.
- Chiappa, Marin Vargas, Huang, and Mathis. Latent Exploration for Reinforcement Learning. NeurIPS 2023. arXiv:2305.20065, 2023.
- Chua, Calandra, McAllister, and Levine. Deep Reinforcement Learning in a Handful of Trials using Probabilistic Dynamics Models. NeurIPS 2018. arXiv:1805.12114, 2018.
- Deisenroth and Rasmussen. PILCO: A Model-Based and Data-Efficient Approach to Policy Search. ICML 2011, pp. 465-472, 2011.
- Feng, Xu, and Liu. MuscleVAE: Model-Based Controllers of Muscle-Actuated Characters. SIGGRAPH Asia 2023. arXiv:2312.07340, 2023.
- Ha and Schmidhuber. World Models. arXiv:1803.10122, 2018, 2018. The NeurIPS 2018 camera-ready is a distinct paper: *Recurrent World Models Facilitate Policy Evolution*, arXiv:1809.01999, whose arXiv comments note substantial text overlap with 1803.10122.
- Hafner, Lillicrap, Ba, and Norouzi. Dream to Control: Learning Behaviors by Latent Imagination. ICLR 2020. arXiv:1912.01603, 2020.
- Hafner, Lillicrap, Norouzi, and Ba. Mastering Atari with Discrete World Models. ICLR 2021. arXiv:2010.02193, 2021.
- Hafner, Pasukonis, Ba, and Lillicrap. Mastering diverse control tasks through world models. Nature 640(8059):647-653, 2025. DOI 10.1038/s41586-025-08744-2, 2025. Preprint: *Mastering Diverse Domains through World Models*, arXiv:2301.04104, 2023.
- Hansen, Su, and Wang. TD-MPC2: Scalable, Robust World Models for Continuous Control. ICLR 2024. arXiv:2310.16828, 2024.
- Janner, Fu, Zhang, and Levine. When to Trust Your Model: Model-Based Policy Optimization. NeurIPS 2019. arXiv:1906.08253, 2019.
- Kawato. Internal Models for Motor Control and Trajectory Planning. Current Opinion in Neurobiology 9(6):718-727, 1999, 1999.
- Kidambi, Rajeswaran, Netrapalli, and Joachims. MOREL: Model-Based Offline Reinforcement Learning. NeurIPS 2020. arXiv:2005.05951, 2020.
- Kim and Lee. FreeMusco: Motion-Free Learning of Latent Control for Morphology-Adaptive Locomotion in Musculoskeletal Characters. SIGGRAPH Asia 2025. arXiv:2511.14205, 2025.
- Marschall, Bund, and Wiemeyer. Does Frequent Augmented Feedback Really Degrade Learning? A Meta-Analysis. E-Journal Bewegung und Training 1:74-85, 2007, 2007.
- Rolf, Steil, and Gienger. Goal Babbling Permits Direct Learning of Inverse Kinematics. IEEE Transactions on Autonomous Mental Development 2(3):216-229, 2010, 2010.
- Salmoni, Schmidt, and Walter. Knowledge of Results and Motor Learning: A Review and Critical Reappraisal. Psychological Bulletin 95(3):355-386, 1984, 1984.

- Schumacher, Haeufle, Büchler, Schmitt, and Martius. DEP-RL: Embodied Exploration for Reinforcement Learning in Overactuated and Musculoskeletal Systems. ICLR 2023. arXiv:2206.00484, 2023.
- Sekar, Rybkin, Daniilidis, Abbeel, Hafner, and Pathak. Planning to Explore via Self-Supervised World Models. ICML 2020. arXiv:2005.05960, 2020.
- Seo, Lee, James, and Abbeel. Reinforcement Learning with Action-Free Pre-Training from Videos. ICML 2022. arXiv:2203.13880, 2022.
- Shadmehr and Mussa-Ivaldi. Adaptive Representation of Dynamics During Learning of a Motor Task. *Journal of Neuroscience* 14(5):3208-3224, 1994, 1994.
- Winstein and Schmidt. Reduced Frequency of Knowledge of Results Enhances Motor Skill Learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 16(4):677-691, 1990, 1990.
- Wochner, Schumacher, Martius, Büchler, Schmitt, and Haeufle. Learning with Muscles: Benefits for Data-Efficiency and Robustness in Anthropomorphic Tasks. CoRL 2022. arXiv:2207.03952, 2022.
- Wolpert and Kawato. Multiple Paired Forward and Inverse Models for Motor Control. *Neural Networks* 11(7-8):1317-1329, 1998, 1998.
- Wolpert, Ghahramani, and Jordan. An Internal Model for Sensorimotor Integration. *Science* 269(5232):1880-1882, 1995, 1995.
- Yu, Thomas, Yu, Ermon, Zou, Levine, Finn, and Ma. MOPO: Model-based Offline Policy Optimization. NeurIPS 2020. arXiv:2005.13239, 2020.
- Yu, Kumar, Rafailov, Rajeswaran, Levine, and Finn. COMBO: Conservative Offline Model-Based Policy Optimization. NeurIPS 2021. arXiv:2102.08363, 2021.